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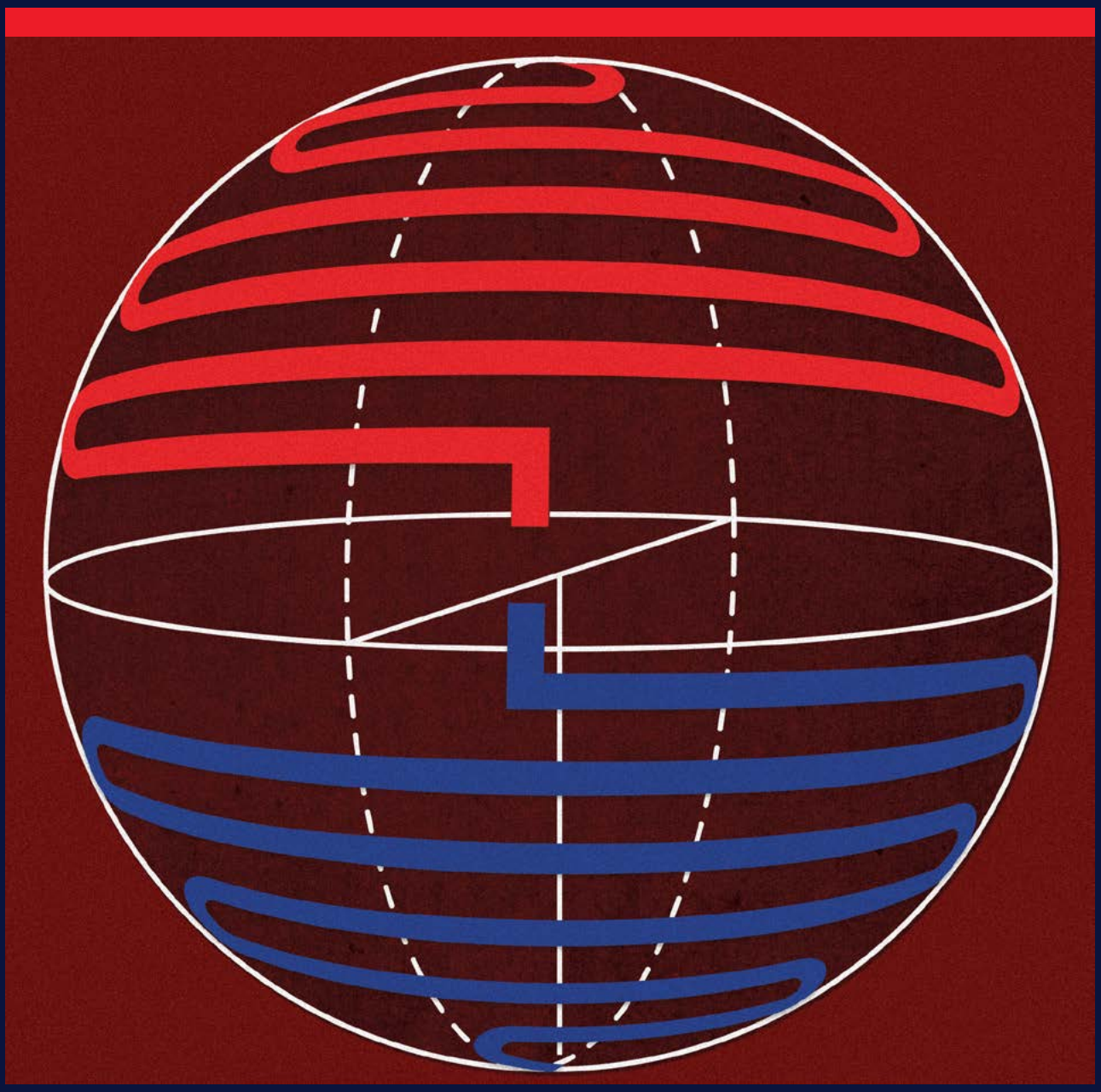
The Quest for Qubits

Assessing U.S.-China Competition in Quantum Computing

Sam Howell



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About the Author

Sam Howell is an adjunct associate fellow at the Center for a New American Security (CNAS) and a military operations and security force assistance analyst on the Afghanistan War Commission. Her research interests include quantum information science, semiconductors, STEM workforce issues, and the use of emerging technologies to enhance human performance. Howell previously worked as an intelligence analyst at a risk consulting firm in Washington, DC. She also worked as a counterterrorism researcher at the National Consortium for the Study of Terrorism and Responses to Terrorism (START) and the Elliott School of International Affairs. She began her career as a staff assistant on Capitol Hill. Howell's commentary and analysis have appeared in outlets including *The Economist*, *Foreign Policy*, Lawfare, *The Diplomat*, and Voice of America. She holds an MA in Security Policy Studies from the George Washington University and a BA in Bioethics and Political Science from Baylor University.

About the Technology and National Security Program

The CNAS Technology and National Security Program explores the policy challenges associated with emerging technologies. A key focus of the Program is bringing together the technology and policy communities to better understand these challenges and together develop solutions.

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Executive Summary

The United Nations General Assembly recently declared 2025 the International Year of Quantum Science and Technology.[1] Quantum information science, a field once exclusive to academic researchers, has rapidly expanded to "include significant private sector development and commercialization."[2] At this particularly opportune moment, science now "stands at the verge of a new quantum revolution."[3]

Amid climate crises, widespread food insecurity, pandemics, and countless other societal challenges, the new quantum revolution promises to benefit nearly every facet of modern life. Quantum technologies, with their unparalleled speed and processing power, could unlock transformative innovations in a range of industries, from energy and agriculture to medicine and finance. At the same time, however, irresponsible actors could use quantum technologies to break encryption, enable mass surveillance, and design new weapons systems. In short, global advancements in quantum technology have profound implications for U.S. economic and national security. The United States needs a long-term strategy to boost its competitiveness in quantum technology and ensure that the technology develops in alignment with U.S. values and interests.

This report takes stock of the global quest to build and scale quantum computers and emphasizes the U.S.-China dynamic. To counter the risk of U.S. adversaries achieving surprise scientific breakthroughs and weaponizing quantum systems, U.S. policymakers must understand the potential power of quantum computers, the state of quantum technology research and development, and the barriers to continued progress. They must also understand what U.S. adversaries hope to achieve in quantum technology and the implications of ceding U.S. leadership in this critical field.

This report arrives at several policy recommendations for the United States to secure its advantages in quantum computing and mitigate any vulnerabilities. The recommendations fall under two particularly pressing policy areas: supply chain management and workforce development.

Supply Chain Management

A well-established quantum technology supply chain with limited dependence on foreign suppliers is critical to U.S. leadership and competitiveness in quantum technology. To foster the development of a secure and reliable supply chain, the White House should:

- *Conduct a comprehensive review of the quantum technology supply chain* to identify key inputs, detect critical nodes and potential vulnerabilities, and assess opportunities to develop additional suppliers across a range of regions.
- *Leverage financial incentives and the Tech Hubs Program to promote domestic production of quantum technology inputs deemed too sensitive to reside outside of the United States.* The White House should leverage the Defense Production Act, Small Business Innovation Research program, Small Business Technology Transfer program, and related tools to support U.S. companies developing key quantum technology inputs, and incentivize them to align research and development efforts with U.S. government priorities.
- *Direct and resource the Quantum Economic Development Consortium to continually monitor U.S. and foreign competitor quantum computing supply chains.* Early detection of emerging vulnerabilities will give policymakers the opportunity to shape the quantum computing supply chain before the United States develops unsustainable dependencies on competitors. Supply chain monitoring will also help illuminate which modalities adversaries are pursuing and how rapidly they are progressing toward fault-tolerant quantum computing.
- *Establish an international forum to coordinate supply chains with U.S. allies and partners,* which will allow like-minded countries to maximize their comparative advantages and facilitate supply chain burden sharing.

Workforce Development

Access to a skilled and diverse quantum workforce is critical to a country's competitiveness in quantum technology. Human talent is required to design quantum technologies, identify quantum applications, and ensure responsible technology use. But quantum expertise is a commodity in short supply and skills gaps could undermine the United States' ability to reach critical development milestones. The U.S. government, private industry, and academia can take several steps to grow and enhance the United States' pool of human talent.

- *The National Quantum Coordination Office should develop processes to understand workforce requirements and skills gaps within the quantum ecosystem.* The United States needs a mechanism to pinpoint human talent vulnerabilities, identify opportunities for reskilling and upskilling initiatives, and assess the capacity of U.S. institutions—including universities, national labs, and private companies—to effectively nurture quantum talent. An annual systematic study of quantum workforce needs and projections will help policymakers, business leaders, and the academic community make more informed decisions about how best to foster a quantum-capable workforce.

- *Congress should establish a national quantum education and workforce development center to harmonize and support existing initiatives to build a more robust pipeline of quantum talent.* A national center will provide a needed venue to coordinate disparate ongoing workforce development programs, share resources and best practices, and connect job seekers to relevant employers.

- *The White House should work with allies and partners to create cross-training and reciprocal research exchange programs with quantum centers worldwide.* No country has exclusive access to the people required to advance state-of-the-art quantum technology. The White House should collaborate with allies and partners so that every country can benefit from readily available quantum expertise. The Biden administration should start by increasing the size of the Quadrilateral Security Dialogue (Quad) STEM Fellowship program and leveraging the budding U.S.-Japan-South Korea trilateral relationship to activate new workforce development initiatives.

- *Universities and quantum companies should work together to develop industry-specific postsecondary education programs,* which will ensure that students gain the practical experience required to fill critical quantum talent gaps immediately upon graduation.

- *Quantum companies should partner with K–12 school systems to develop quantum curricula and experiential learning opportunities for students.* Such engagement will help teachers gain confidence teaching quantum principles and increase student exposure to and interest in quantum technology.

- *Quantum companies should partner with state and local governments to expand quantum education and professional development opportunities to individuals in rural areas.* Quantum apprenticeship programs, accelerated skill development programs, online certification programs, or similar initiatives will allow individuals in rural areas—a sizable untapped talent pool—to hone their skills and advance their earning and promotion potential, ultimately contributing to a more robust and diverse quantum workforce.

Introduction

The United States and the People's Republic of China are locked in a long-term, strategic contest for geopolitical power. One critical and emerging technology area—quantum information science (QIS)—is set to play an outsized role in determining which country prevails.

China increasingly aims to reshape the long-standing U.S.-led international order in its own image and is developing the economic, diplomatic, and military capabilities to do so. The gap in national strength between the United States and China is narrowing rapidly. For the first time since the Cold War, the United States risks losing its position as the global superpower.

Technology is a key battleground of U.S.-China competition. Critical and emerging technologies—including QIS, artificial intelligence, biotechnology, and semiconductors—are tools that help countries promote their interests and maximize their global competitiveness. As National Security Advisor Jake Sullivan acknowledged in a 2022 speech, "Advancements in science and technology are poised to define the geopolitical landscape of the 21st century."[4] The United States cannot prevail in strategic competition without first establishing leadership in emerging technologies.

Maintaining an edge in QIS, a technology area with tremendous economic and military potential, is especially important to U.S. technological and global leadership. U.S. government officials across multiple administrations have identified QIS as an area that demands attention over the coming decade.[5] They understand that competitiveness in quantum technology could significantly enhance overall national competitiveness. The countries with the best quantum technologies could also have the best pharmaceuticals, batteries, fertilizers, intelligence collection, and weapons systems.

Quantum computing is a subset of QIS that harnesses the laws of quantum mechanics to solve large and complex problems that are otherwise intractable on classical computing counterparts.

Quantum technologies promise to support transformative advancements in industries like healthcare, finance, and agriculture.[6] Quantum computers could simulate interactions between complex molecules, paving the way to more effective medications, diagnostics, and vaccines. They could also interpret financial data and model trends, contributing to dramatically improved portfolio optimization, risk analysis, fraud detection, and capital allocation. Quantum computers even have a role to play in mitigating climate change and fostering sustainability. They could accelerate improvements in electric battery technology, unlock more eco-friendly fertilizer manufacturing processes, and help solve power grid optimization problems.[7] Nearly every facet of modern life stands to benefit from the unmatched power of quantum computers.

But quantum computers could become a double-edged sword. Countries that harness the power of quantum computers for good might simultaneously use the technology for more nefarious purposes. Quantum systems could help improve the accuracy and lethality of militaries and intelligence services, making it easier for governments to identify and neutralize targets or surveil and repress specific populations. Quantum systems could also help countries design new weapons, overwhelm adversaries' defensive capabilities, or rapidly analyze data across multiple domains of warfare.[8] Many emerging technologies possess such dual-use applicability, but quantum technologies are exceptionally challenging to manage—quantum is a relatively new and rapidly evolving technology whose full potential has yet to be fully explored or even conceived. The first-mover advantages of quantum technologies are thus significant. The first country to master the process of building a useful quantum computer will possess not just a powerful tool, but technical know-how that is extremely difficult to cultivate.

The first country to develop, scale, and integrate quantum technologies could attain an upper hand in establishing market dominance, setting technology standards, and crafting frameworks for quantum governance. The global leader in QIS will gain a set of tools that could be used to crack encryption methods and threaten the security of adversaries' corporate, military, and government infrastructure. Quantum computing could also bestow new capabilities to shape ethical norms—the technology holds the potential to exacerbate global socioeconomic divides, breach individuals' privacy and security, and contribute to job displacement.[9] In other words, the country leading in QIS will significantly influence whether quantum technologies are used to promote and protect democratic values or undermine them.

There are three primary subsets of quantum technologies: quantum computing, quantum sensing, and quantum communications. This report focuses on

quantum computing, the subset of QIS that has received the largest share of public and private funding in the United States thus far, reflecting its perceived importance to U.S. national security.[10] It is also the subset in which the United States arguably faces the fiercest competition. Quantum computing has the most competitors among the three QIS subsets, the largest estimated market, and the most funding globally. At least 17 countries have national strategies to support quantum computing research and development (R&D), including the United States and China.[11] According to most estimates, China outspends the United States on quantum technology R&D by approximately $13.4 billion.[12]

This report explores how the United States can maintain and advance its competitiveness in quantum computing. It provides an overview of QIS and the current state of quantum computing R&D. It also compares the United States' and China's quantum computing priorities and capabilities, as well as their capacity for continued innovation. Thereafter, the report highlights two key policy areas—supply chain management and workforce development—that demand policymakers' immediate attention. U.S.-China competition in quantum computing raises several pressing policy concerns, including questions about export controls, outbound investment restrictions, standard setting, and international governance frameworks. However, the United States cannot effectively address these questions without first establishing a resilient quantum supply chain and fostering the workforce required to remain at the cutting edge of this fast-moving field. The report concludes with recommendations to maximize U.S. advantages in these areas to ensure sustained American quantum competitiveness.

Technology Primer: What Is Quantum?

Two major advances of the 20th century—quantum mechanics and information technology—gave rise to the field of QIS in the 1980s.[13] Quantum mechanics is a subfield of physics that describes the behavior of nature at the atomic and subatomic level.[14] Information technology involves the use of so-called classical computers, digital communication, and related devices to solve problems on a broad scale.[15] QIS combines the two fields by leveraging "the laws of quantum physics for the storage, transmission, processing, and measurement of information."[16] QIS will theoretically provide the capability to overcome the physical limits of classical information systems and technologies, unlocking new discoveries and innovations in a range of industries.

Quantum computing is a subset of QIS that harnesses the laws of quantum mechanics to solve large and complex problems that are otherwise intractable on classical computing counterparts.[17] Quantum computers take advantage of several quantum-mechanical phenomena that distinguish them from classical computers. The foundation of classical computing is binary *bits* that switch on and off and can only exist in states of either zero or one. Quantum computers are based on *qubits*, which can exist in multiple combinations of states. This phenomenon is referred to as *superposition*.[18] Superposition makes quantum computers significantly more powerful than classical computers. Because qubits can represent zero, one, or any combination of the two, the number of possible states for a qubit far exceeds the possibilities for binary bits, giving quantum computers greater capacity to absorb, explore, and process information.

However, qubits can only hold superposition until an observation, measurement, or undesired environmental disturbance occurs, at which point they collapse into a single state. Each quantum state maintains a certain probability ranging between zero and one. *Quantum interference*, which refers to the way that wavelike quantum particles in a superposition of multiple states engage with each other, is one way to affect probability amplitudes and bias the outcome of an observation or measurement toward a particular state.[19] Interference is used to control quantum states, perform computations, and detect and correct computational errors. It can also support significant improvements in a quantum computer's computational speed.[20]

Visualizing a "quantum coin" illuminates the concepts of superposition and interference. A traditional coin only exists in one of two possible states—heads or tails. But a quantum coin can exist in a superposition of these classical states. When the quantum coin is not being observed or measured, it can be any particular combination of both heads and tails. Once one measures the quantum coin—such as observing the result of a coin toss—the quantum coin collapses into one of its two states, heads or tails.[21] Interference allows the observer to increase the likelihood that the coin lands on their desired outcome. In other words, interference helps program quantum systems so that correct answers are more likely and incorrect answers are less likely.[22]

Entanglement is another unique quantum phenomenon that affords quantum computers unparalleled power and speed. Unlike digital bits that have

independent states, entangled qubits interrelate in such a way that the properties of one cannot be described independently from the properties of the others, even when separated by distance. Entangled qubits cohere with each other and function as a connected system or unit. One can measure the state of a qubit and know the state of the others in the system, even without measuring them directly.

Assume, for example, that two particles are in a superposition of red and blue under conditions in which, if one is red, the other must be blue. The concept of entanglement means that measuring the color of one particle fixes the color of the other, regardless of any space between the two. Like superposition, entanglement enables quantum systems to process exponentially higher volumes of information than their classical counterparts and gain exponentially more power with the addition of more qubits that can interact and communicate with each other. This means that quantum computers can solve a narrow subset of complex problems that are beyond the capacity of classical computers.

Technologies that rely on these quantum phenomena already exist, including lasers, the Global Positioning System (GPS), and magnetic resonance imaging (MRI).[23] But existing technologies do not require a significant amount of control over quantum particles and only leverage quantum effects to a limited degree.

Researchers and engineers must address several obstacles to build more robust quantum technologies. One such obstacle is *quantum decoherence*, which refers to quantum systems' fragility and sensitivity to external disturbances such as variation in magnetic fields or radiation from nearby objects. Decoherence causes quantum particles to fall out of superposition and entanglement and lose their utility.[24] The QIS field broadly aims to "advance the theory and experimental capabilities needed" to increase the amount of time that particles can maintain superposition and entanglement (referred to as coherence times), control quantum particles, and leverage quantum phenomena for new tasks and capabilities.[25]

Quantum Computing Applications

Quantum computers lack general-purpose functionality and will not replace classical computers for a wide range of classical computing jobs. Classical computers will continue to outperform quantum computers at simple, straightforward tasks like arithmetic operations, as well as tasks like web browsing, email communication, and word processing.[26] Quantum computers are best suited for specific applications. They are expected to be particularly good at solving optimization, simulation, and prime factorization problems.

A few key quantum algorithms—which define the steps required to complete computations—enable these applications.[27] Shor's algorithm, which finds the prime factors of an integer exponentially faster than classical algorithms, supports prime factorization problem-solving.[28] The algorithm could thus enable quantum computers to break the widely implemented public-key encryption systems near-universally relied on for secure data transmission and protection. The algorithm could also undermine the security of most blockchain protocols, which rely on the inability of classical computers to factor large numbers.[29] Grover's algorithm, which rapidly searches large databases and identifies ideal solutions from a set of available alternatives, supports optimization problem-solving. Quantum computers running Grover's algorithm could potentially help maximize efficiency and cost savings across a wide range of industries.[30]

Quantum computers are also well-suited to simulate quantum-mechanical effects and illuminate quantum interactions within large, complex systems. This feature could expedite scientific learning, research, and discovery in areas like biochemistry and advanced materials.[31] Newer quantum algorithms indicate that quantum

Key Concepts

Qubit: The basic unit of information in quantum computing.

Superposition: Qubits can represent zero, one, or any combination of the two states.

Entanglement: The properties of qubits cannot be described independently of one another, even when separated by distance.

Quantum Interference: The ability of one particle's wavelike action to diminish or amplify the action of other particles. In quantum computing, this wavelike interaction between particles can be used to influence the probability of outcomes when a quantum state is measured.

Quantum Coherence: Qubits' ability to maintain superposition and entanglement during computation.

Noise: External disturbances—such as variation in magnetic fields, radiation from nearby objects, or cross talk between particles—that cause qubits to decohere or fall apart.

computers could also help solve various mathematical problems—such as linear differential equations, least-square curve fitting, and matrix inversions—far faster than classical computers. The Harrow-Hassidim-Lloyd quantum algorithm, for example, performs rapid linear algebra calculations and could be used to develop more robust machine learning systems or overcome complex engineering challenges.[32]

In short, quantum computers could solve highly complex problems that classical computers cannot, opening significant real-world opportunities—even if they will not replace classical computers for a wide range of compute problems.[33] It is still unclear, however, whether quantum computers can solve these highly complex problems at a reasonable cost. Grover's algorithm, for example, may not provide enough of an efficiency advantage to offset the time and resources required to perform error correction, a process described in subsequent sections that gives quantum computers the capacity to execute the algorithm.[34] But quantum researchers and engineers are working to overcome these overhead costs, and the benefits are already apparent in the aerospace, automotive, and port logistics industries.

The U.S. aerospace company Boeing is exploring the use of quantum computers to design ply composites, the fundamental building blocks of wings, fuselages, and other critical aerospace components. Ply design problems, which require "finding the optimal way to stack layers of material on top of one another" and can involve up to 100,000 variables, are too large and complex for classical computers.[35] Boeing's routine approach to ply design thus involves a time and resource-intensive process of breaking problems into smaller pieces that a classical computer can digest and patching together disparate results.

Quantum computers could help streamline the process. In 2023, Boeing researchers successfully ran a ply composite problem involving 40 variables on an IBM quantum computer. The experiment demonstrated that "a single qubit could handle three times the information load of a classical bit and represent three times as many variables."[36] With additional maturation, quantum computers could expedite and simplify Boeing's design process by solving ply composite problems in a single computation. Boeing is already exploring additional quantum computing applications as well, including the use of quantum computers to develop advanced corrosion-resistant chemicals.[37]

In the automotive industry, Mercedes-Benz is exploring the use of quantum computers to aid in the design of lithium-ion (Li-ion) batteries, an essential component of electric vehicles, energy storage systems, and other green technologies. Classical computers lack the capacity to efficiently find solutions to Li-ion electrolyte optimization problems, a key feature of battery design. Quantum computers can use superposition and entanglement to explore possible chemical reactions in liquid Li-ion electrolytes at the molecular level and identify possible designs. They can then leverage interference to amplify optimal designs and diminish suboptimal ones.

Quantum computers' computational power and speed could improve understanding of electrochemical reactions and pave the way to higher-performing car batteries.[38] Instead of testing hundreds of different chemicals for their performance in a battery, quantum computers could simulate their performance more efficiently and reduce the number of physical experiments in chemistry laboratories. Quantum computers could be used to solve a series of other optimization problems in the automotive industry as well. BMW, for example, is investigating the use of quantum computers to determine the best locations to install electrical charging stations, find more efficient ways to purchase BMW components, and increase aerodynamic performance and safety.[39]

The Port of Los Angeles has also used quantum computing to improve infrastructure capacity, efficiency, and supply chain resilience at Pier 300, the nation's largest facility for handling shipborne cargo. The Port deployed the Hyper Optimization Nodal Efficiency (HONE) quantum-powered AI engine in July 2020 to create a "digital twin" of the terminal, identify key inefficiencies, and simulate possible solutions. HONE has contributed to "dramatically streamlined port operations" across a number of key productivity performance indicators. The terminal uses nearly 40 percent less rubber-tired gantry crane resources for the unloading process since deploying HONE. The cranes have also increased the number of deliveries by more than 60 percent and trucks arriving at the terminal spend approximately 10 minutes less receiving their payload.[40]

Barriers to the Quantum Breakthrough: Error Mitigation and Correction

Though quantum computing already demonstrates great potential, it is a nascent technology that requires additional maturation to deliver on its full economic and military potential. Quantum computers available today are noisy intermediate-scale quantum (NISQ) computers—they are prone to considerable error rates and relatively small in scale, which limits their applicability to exploration and discovery.[41] While NISQ computers

can perform a few simple tasks, quantum computers—and qubits specifically—must reach a higher level of complexity to solve real-world problems successfully and accurately. Fault-tolerant quantum computers that can compensate for qubits' inherent fragility and respond to failures in quantum processors are key to executing the most advanced quantum algorithms. *Quantum error mitigation* and *quantum error correction* are tools to offset the weaknesses of NISQ computers and unlock fault tolerance.

Quantum error mitigation aims to address errors that occur during algorithmic execution and impact the measurement of a quantum computer's output. Rather than fixing a specific problem, error mitigation simply aims to increase the stability of the computational environment. One approach to quantum error mitigation—which includes "randomized compiling, measurement-error mitigation, zero-noise extrapolation, and probabilistic error cancellation"—can involve "running many slightly different versions of a target algorithm, combining the results to extract the correct answer" from a series of flawed outputs, and modifying the original algorithm to lessen the effect of errors.[42] Error mitigation strategies vary in their levels of accuracy, and each approach has tradeoffs. An algorithm must run hundreds of times with different configurations to derive maximum benefit from most error mitigation strategies, for example, which increases computing time and costs.[43]

Quantum error correction (QEC) aims to build redundancies so that a quantum system produces accurate outcomes even if a few qubits experience errors. QEC involves encoding quantum information across multiple physical qubits to form a logical qubit. These qubits monitor the quantum processor and detect when errors occur, without disturbing the information stored in individual qubits. A correction code is then deployed based on the type of error. The most common errors include *phase flip errors*, in which a qubit loses its phase information, *bit flip errors*, in which qubits depolarize into other states, and *gate operation errors*, in which quantum gates damage qubits.[44] QEC is a "feedback stabilization process that uses indirect measurements to gain the information required to correct these errors."[45]

QEC is a necessary step toward fault-tolerant quantum computing, but it is challenging to implement. The quantum threshold theorem indicates that quantum hardware must reach a minimum error rate before it can benefit from error correction.[46] Current estimates suggest that the error rate for the minimum unit of quantum computation—a quantum gate—must be below 1 percent to run the surface code, one of the most promising QEC codes.[47] Researchers are progressing toward hardware that meets threshold requirements, but building systems that can perform entangling quantum operations with high fidelity remains a central engineering challenge.[48]

Further, some leading error-correcting codes require more qubits than today's quantum machines can reasonably produce. The ratio between the number of physical qubits required to support a single logical qubit varies depending on the type of hardware, the type of algorithm, and the amount of noise a quantum system endures, but a large number of qubits could be required to create one useful error-corrected logical qubit.[49] Google, which is at the leading edge of QEC development, announced that it created a logical qubit from 49 physical qubits in February 2023.[50] The Google team admitted that while the "error correction demonstration shows a clear path for scaling up the size of quantum computing systems, it still falls well short of the error correction rate needed to make quantum computing practical."[51]

Quantum computers will not reach maximum utility until researchers overcome qubits' fragility and susceptibility to errors. To unlock the most advanced applications of quantum computing, quantum systems must be large and scalable with the ability to manage a lot of qubits—potentially up to one million—in a single computation.[52] But scale alone is insufficient. Fault-tolerant quantum computing also requires high-quality, error-corrected qubits that can maintain coherence for extended periods of time. This means that the number of qubits in a quantum system is just one important metric for assessing performance. The system's error and fidelity rates matter too.

Quantum Computing Modalities

As countries around the world race to develop the first full-scale quantum computer, governments and companies have placed bets on which quantum hardware architecture will prove most useful or commercially viable. Quantum computers have several possible designs and there are multiple modalities under development. The approach to developing qubits, the building block of the quantum computer, distinguishes one modality from another.[53]

A quantum modality or hardware architecture must do three things to deliver real-world impact: 1) create well-characterized qubits, 2) allow qubits to be controlled and measured, and 3) correct or mitigate errors.[54] The system must also do these things at scale, meaning it must manage a lot of qubits in a single computation.[55] It is not yet clear which modality will be the first to meet

these criteria, and it is possible that multiple modalities could eventually check all the boxes.

No modality is flawless, but several demonstrate strong potential. The most promising architectures at this stage of development include superconducting qubits, trapped ions, photonics, and neutral atoms.[56] The United States and China maintain capabilities in each of these four areas, and both countries aim to be the first to determine which modalities meet the criteria for impactful quantum computing. The subsequent section offers a brief overview of each modality to illuminate the differences between them and the overall state of the quantum computing development race.

SUPERCONDUCTING QUANTUM COMPUTERS

Superconducting quantum computers rely on qubits developed from electrical circuits that have been cooled to near–absolute zero temperatures. Electrical circuits are composed of metals that transmit an electric current. When cooled to low temperatures, they become superconducting and quantum mechanical, creating a solid-state qubit. They conduct a current without resistance, do not lose energy to the environment, and can occupy specific energy states. Physical operations—such as the application of microwave pulses or magnetic flux pulses—help control and measure superconducting qubits.

Superconducting qubits have fast operation times and complete computations faster than several molecular qubit architectures. Additionally, many of the fabrication techniques used to build superconducting qubits are the same as those used to build classical technologies, meaning that they have existed for decades and undergone continual refinement. Superconducting qubits are thus relatively easy to construct and could be more scalable than qubits that require novel techniques. Engineers also have significant control over the design

Advantages and Disadvantages of the Leading Quantum Computing Modalities[57]

Modality	Advantages	Disadvantages	Leading Companies
Superconducting Qubits	▪ Operation speed ▪ Well-developed fabrication techniques ▪ Artificially manufactured—high degree of control over design ▪ Circuit logic similar to classical computing	▪ Quick to decohere ▪ Dependent on cryogenic cooling ▪ Large infrastructure and power requirements ▪ Artificially manufactured—no two qubits are exactly alike	▪ IBM ▪ Google ▪ Atlantic Computing ▪ Rigetti Computing ▪ Origin Quantum Computing ▪ Tencent
Trapped Ions	▪ Long coherence times ▪ Low error rates ▪ Naturally occurring—identical qubits ▪ Can potentially operate at room temperature ▪ Strong connectivity across qubits	▪ Slow gate times, measurement, shuttling, and cooling ▪ Immature fabrication techniques ▪ Potential for crosstalk between qubits	▪ Quantinuum ▪ IonQ ▪ Qike Quantum ▪ Alpine Quantum Technology
Photonics	▪ Qubits resistant to interference ▪ Fast gate times ▪ Low cryogenic cooling demands ▪ Well-developed fabrication techniques ▪ Leverages microchip fabrication techniques	▪ Not all-purpose machines ▪ Lack of connectivity between photons ▪ Difficult to entangle and scale	▪ Xanadu Quantum Technologies ▪ Orca Computing ▪ PsiQuantum ▪ TundraSystems Global ▪ Quandela ▪ Bose Quantum ▪ TurningQ ▪ Tencent
Neutral Atoms	▪ Strong coherence and connectivity ▪ Low error rates ▪ Naturally occurring—identical qubits ▪ Can potentially operate at room temperature ▪ No complex electrical wiring required	▪ Challenging to scale ▪ Need for more advanced laser technologies ▪ Slower processing times ▪ Nondestructive measurement is challenging	▪ Atom Computing ▪ Pasqal ▪ QuEra

of superconducting qubits, which could enable more complex and tailored computational operations.[58]

At the same time, no two superconducting qubits are exactly alike because they are manufactured, unlike counterpart modalities like neutral atoms, which build qubits out of naturally occurring individual atoms. Superconducting quantum computers thus require calibration that could become complicated as the size of systems increases. Superconducting qubits also decohere—or fall out of alignment with one another—quickly, resulting in information losses. This means that superconducting quantum systems may require more qubits, operations, and error correction to perform accurate computations.[59]

Superconducting qubits' dependence on cryogenic systems—the mechanism by which qubits are cooled to near–absolute zero temperatures—is another disadvantage. Cryogenic systems are expensive, only available from a limited number of providers, and reliant on helium, a scarce commodity.[60] The compressors used to precool cryogenic systems also consist of magnets containing rare-earth elements that are largely located and processed in China, presenting another potential supply chain challenge.[61] Despite their pitfalls, superconducting qubits form the foundation of several leading-edge quantum computers, including IBM's Condor and Google's Sycamore.[62]

ION-TRAP QUANTUM COMPUTERS

Ion-trap quantum computers rely on qubits developed from ions or charged atoms. Electromagnetic fields "trap" or suspend ions in free space and align their electronic states. Trapped ions can be spin up, spin down, or exist in a superposition of these two states. The application of optical and microwave signals to suspended ions enables manipulation of their states and induces coupling between them.[63] Such coupling affords ion-trap quantum computers infinitely more computational speed and processing power than conventional computers.

Stability is a key advantage of the qubits used in ion-trap computers. The complex array of electric and magnetic fields in trapped-ion quantum computers absorb random motions and noise from the environment. Trapped ions thus demonstrate longer coherence times and lower error rates than superconducting qubits. Ions are also naturally occurring, which means they are identical, interchangeable, and can interact easily with each other.[64]

Trapped ions can also operate at room temperature. Cooling is required to maximize the performance of trapped ions, but they demand less cooling capacity than superconducting qubits. While superconducting qubits must be cooled to nearabsolute zero, trapped ions perform sufficiently at approximately four degrees above absolute zero, a significant difference at this level despite the seemingly small gap in temperature.[65]

Slow gate times—which refers to the amount of time required to perform an operation on a quantum state—are the primary disadvantage associated with trapped ions. Trapped-ion quantum computers require more time to complete operations than superconducting quantum computers. It also takes a lot of time to cool, move, and measure trapped-ion quantum computers. Further, the technology involved in creating ion traps—such as the vacuum used to hold the ions—is somewhat less mature than the technology used to fabricate superconductors. Trapped-ion quantum computers could thus be more difficult to scale.[66]

PHOTONIC QUANTUM COMPUTERS

Photonic quantum computers rely on qubits developed from particles of light, which can vibrate in different directions or polarized states. Photonic systems fire laser beams at a finely polished piece of glass, causing beams to break into two and vibrate in unison. These coherent beams reflect to a common point at which the photons entangle with one another, and the resulting beam exists in a superposition of the entangled photons' states, creating a qubit.[67]

There are several benefits associated with photonic quantum computing. Unlike noise-sensitive electrons, photons are resilient against disturbances from the environment because they are not charged. Photonic quantum computers thus require fewer qubits for robust fault-tolerant operations than electron-based systems. Photons also travel at the speed of light. This means that they can theoretically complete more operations before they decohere, or lose their information.[68] As a result, photonic quantum computers may be able to run advanced algorithms faster than other modalities.

Further, photonics-based systems can operate at room temperature. The process of generating photonic qubits may require cryogenic cooling, but the complete photonic quantum processing unit is not cryostat-dependent, lowering the complexity and cost associated with completing computations. Similar to superconducting quantum computers, photonic quantum computers also rely on well-developed technologies that are already used in telecommunications infrastructure and fiber optics, which lowers the engineering hurdles associated with building scalable systems.[69]

However, photonic computers are not all-purpose machines that can be programmed for instant calculations. Photonic computers historically involve a collection of mirrors and beam splitters that must be rearranged for each problem and consume a lot of space, so their primary applications may be narrow due to time and resource constraints. Much of photonic quantum computing is now based on chips using photonic wavelengths and other fabricated elements, which reduces the number of moving parts. But photonic computers remain bulky relative to other modalities. Photons also do not easily interact with each other, making it difficult to build increasingly complex qubits, and are sensitive to loss, which degrades performance.

NEUTRAL ATOM QUANTUM COMPUTERS

Neutral atom quantum computers rely on qubits developed from atoms, which consist of a positively charged nucleus surrounded by negative electrons. Atoms from the first two groups of the periodic table—such as rubidium, caesium, and strontium— are most commonly used to develop neutral atom qubits because their electronic structures are well-suited for cooling and trapping. Neutral atom quantum computing involves the manipulation of electrons' energy levels, which can be employed as qubit states.[70]

The core of a neutral atom qubit consists of a vacuum cell containing a geometric arrangement of atoms. Strong magnetic fields and lasers are applied to the atoms to slow their movement, cool them to near–absolute zero temperatures, and form an atom cloud. The atom cloud is then moved into a chamber where optical tweezers, or highly focused laser beams, move the atoms into specifically designed arrays. Once an array is filled in an organized grid, laser pulses transform atoms into qubits by adjusting their nuclear spin. Exciting atoms to the highly energized Rydberg state, in which their electrons orbit the nucleus at a much greater distance than usual, facilitates interaction with nearby atoms and entanglement, enabling quantum computations.[71] Once a quantum circuit is complete, the results of the computation are read out optically and qubits can be reinitialized to run new computations.[72]

Strong coherence and connectivity are the key advantages associated with neutral atoms. Relative to other qubit modalities, neutral atoms retain quantum information for long periods of time and can easily interact with each other, allowing for many successive quantum logic operations.[73] They also operate at room temperature, removing the need for cryogenic cooling. Because the functions to control neutral atom qubits are mediated by light propagating through free space, neutral atom quantum computers reduce the need for complex electrical cables and wiring as well.[74]

The primary challenge associated with neutral atom quantum computing is speed—it takes a long time to move atoms, measure atoms, and replenish atoms after they are measured.[75] The tradeoff between the size of neutral atom systems and qubit coherence presents another challenge—increasing the size of a neutral atom system also increases the complexity and time required to construct organized grids of atoms that can stick together. Scientists and engineers may need to improve register preparation, gates, and measurement protocols and develop more advanced laser technologies before neutral atom quantum computers become sufficiently scalable.[76]

Takeaways

It is not yet clear which hardware platforms will enable fault-tolerant quantum computing—the level of performance needed to run complex quantum computations and build useful quantum computers—but superconducting qubits, trapped ions, photonics, and neutral atoms demonstrate the greatest potential at this stage of development.[77]

Given the uncertainty surrounding which hardware architecture will "win" the race to deliver the first fault-tolerant quantum computer, countries that aim to enhance their competitiveness in quantum computing would be wise to fund broad and diverse R&D projects. Governments should devote funds to promising hardware architectures, as well as novel ones that could emerge as unanticipated contenders in the race to build a quantum computer. This distributed approach to R&D will help ensure that countries have technical expertise and experience with whichever quantum computing modalities prove most useful. However, once it is clear which modalities can produce useful quantum systems, a more targeted and concentrated approach to funding may be required to achieve scale and commercialization.

The following section evaluates the United States' and China's respective quantum computing ecosystems. It aims to determine which country currently maintains a lead in quantum computing research, development, and commercialization. It also assesses each country's capacity and potential for continued innovation. The comparative strengths and weaknesses illuminated in this analysis inform subsequent policy recommendations to promote U.S. competitiveness in quantum technology.

A U.S.-China Comparison

Relevant U.S. and Chinese strategy documents indicate that both countries view quantum as a strategically significant technology with the potential to dramatically enhance the economic and military aspects of national power. Though the United States and China have different quantum technology priorities, both ultimately aim to reach the cutting edge of quantum technology R&D.

The United States plans to advance its quantum competitiveness by pursuing a science-based approach to technology development, supporting private industry, and building a quantum workforce to meet the needs of a growing field. China intends to achieve dominance by pushing the boundaries of existing research, developing indigenous quantum capabilities, and eliminating reliance on foreign technology components and materials. Several key metrics—including total R&D expenditures, patents, and research publications—suggest that the U.S. approach is superior. The United States currently maintains an edge over China in the pursuit of scalable, fault-tolerant quantum computing. But this lead is far from secure. China clearly intends to surpass the United States in quantum computing and is garnering the resources and know-how to do so.

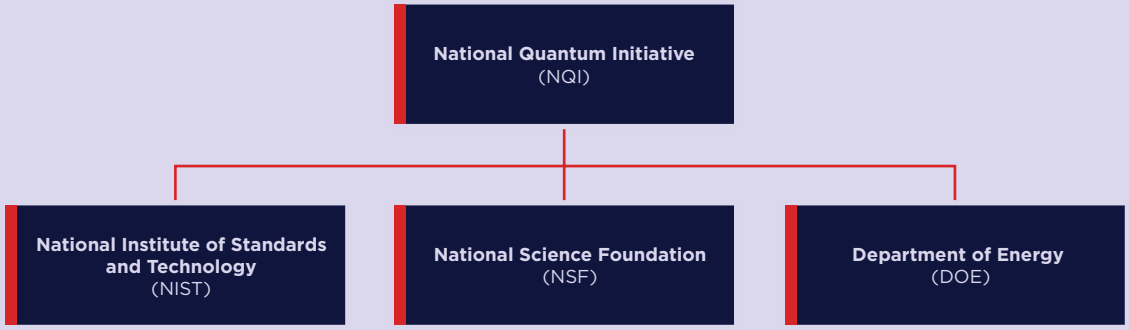
The United States' Quantum Technology Strategy

The United States' quantum technology efforts rest on two policies: the National Strategic Overview for Quantum Information Science (National Quantum Strategy) and the National Quantum Initiative (NQI) Act. Several additional efforts—including the QIS Workforce Development National Strategic Plan, the Quantum Frontiers Report, and National Security Memorandum-10 (NSM-10)—supplement the National Quantum Strategy and the NQI to advance U.S. global leadership in QIS.

The National Quantum Strategy provides the United States' overarching vision and framework for quantum technology R&D. The strategy articulates three primary objectives—the United States aims to 1) pursue a science-based approach to quantum technology development focused on identifying and solving "problems whose solutions enable transformative scientific and industrial progress"; 2) enhance national competitiveness by "accelerating technology development toward useful economic and mission applications of QIS and working with international partners, while also protecting national security"; and 3) "build a quantum-smart and diverse workforce to meet the needs of a growing field."[78] The National Quantum Strategy recommends advancing these objectives by focusing on six areas: science, workforce, industry, infrastructure, security, and international cooperation.

The NQI, initially enacted in 2018, articulates the United States' plan to execute the National Quantum Strategy and serves as a blueprint for quantum-related activities in the United States.[79] The NQI authorizes three agencies—the National Institute of Standards and Technology (NIST), the National Science Foundation

Primary NQI Agencies



(NSF), and the Department of Energy (DOE)—to "establish centers and consortia and carry out new programs to foster QIS R&D."[80] Despite its specificity to NIST, NSF, and DOE, the NQI is a whole-of-government program calling for coordination of QIS R&D efforts across the federal government, as well as with industry and the academic community.[81] Twenty-three federal agencies were involved in the NQI as of April 2022.[82]

The National Quantum Coordination Office (NQCO), located within the White House Office of Science and Technology Policy (OSTP), is responsible for overseeing the NQI program and supporting its implementation.[83] The NQI Act also legislates some oversight and advisory responsibilities to the National Science and Technology Council (NSTC) Subcommittee on QIS (SCQIS), the NSTC Subcommittee on the Economic and Security Implications of Quantum Science (ESIX), and the National Quantum Initiative Advisory Committee (NQIAC).

Two additional pieces of legislation—the Creating Helpful Incentives to Produce Semiconductors (CHIPS) and Science Act of 2022 (CHIPS Act) and the National Defense Authorization Act (NDAA) for fiscal year (FY) 2022—broadened the scope of the NQI Act. The CHIPS Act authorized quantum networking infrastructure R&D, the development of quantum networking and communication standards, the establishment of a program to facilitate researchers' access to quantum computing resources, and the integration of QIS into national education curricula. It also included QIS in the NSF Directorate for Technology, Innovation, and Partnerships, "as well as in existing federal scholarship programs."[84] The FY 2022 NDAA further amended the NQI Act to include the ESIX.[85] The FY 2019 and FY 2020 NDAAs also contain quantum-relevant legislation, including authorization for the Department of Defense (DoD) to conduct and support quantum technology R&D, but do not directly amend the NQI Act.[86]

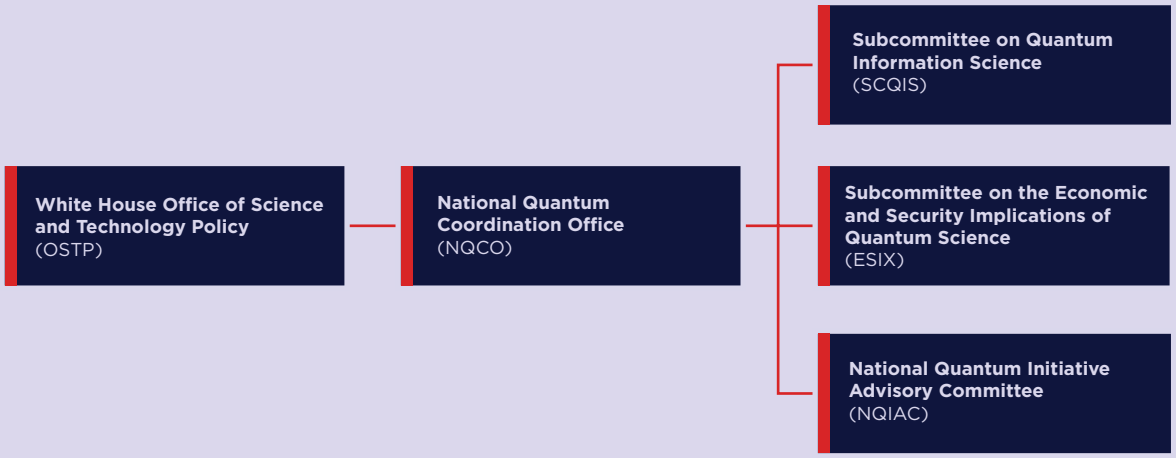
In summary, quantum activities in the United States are informed and guided by the National Quantum Strategy, the NQI, and supplementary documents and legislation. The United States takes a whole-of-government approach to quantum R&D, with robust support from private industry and academia.

PUBLIC SUPPORT FOR QUANTUM COMPUTING

In line with the National Quantum Strategy and the NQI, the U.S. government has made "substantial and sustained investments in fundamental QIS R&D."[87] U.S. government funding aims to foster innovation and exploration of several quantum technology modalities and applications. Federal investments in quantum technology are thus broad and diverse, spanning several different agencies and departments. The primary drivers of QIS R&D spending include the three agencies targeted by the NQI Act—NIST, NSF, and DOE—as well as the DoD, the Department of Homeland Security, and NASA. A fuller list of the entities involved in quantum R&D in the United States is included on page 14.

The United States' distributed funding strategy, coupled with the fact that the United States lacked a designated body to track annual spending on QIS R&D

NQI Oversight Bodies



until the creation of the Subcommittee on Quantum Information Science (SCQIS) in 2019, makes it difficult to ascertain total U.S. federal investment in QIS.[88] It is clear, however, that the NQI has played a significant role in spurring federal interest and investment in QIS. R&D budgets for QIS have roughly doubled since FY 2019. According to the SCQIS, agencies reported actual QIS R&D budget expenditures of $449 million in FY 2019, $672 million in FY 2020, and $855 million in FY 2021, followed by $918 million of enacted budget authority in FY 2022. The Biden administration requested budget authority of $844 million for QIS R&D in FY 2023.[89]

Federal funding for QIS R&D is spread across multiple quantum technology component areas including quantum sensing and networking, but quantum computing has received the largest share of public funding in the United States since at least FY 2021, reflecting its perceived importance to U.S. national security.[90] A 2022 report by McKinsey & Company indicates that 57.6 percent of U.S. government funding for quantum technology has gone toward quantum computing, while quantum sensing and quantum communications have received 28.5 percent and 13.9 percent, respectively.[91] Prior studies suggest that quantum computing has a larger number of distinct federal funding sources than other quantum technology component areas and that federal agencies—particularly defense and intelligence agencies—fund the majority of quantum technology publications in the United States.[92]

Prior studies also indicate that it is difficult to assess whether U.S. government funding is sufficient to accelerate U.S. quantum technology innovation and maintain U.S. competitiveness for three reasons: 1) the U.S. quantum technology ecosystem lacks universal metrics or a clear roadmap by which to measure progress; 2) quantum technologies remain in their infancy, and there are no explicit examples of a dramatic breakthrough that can be directly connected to a funding surge; and 3) there are few "comprehensive measures of how U.S. QIS research has changed over time."[93] However, there is "widespread consensus" that additional funding is required to continue driving progress. In March 2023, the NQIAC noted in recommendations to Congress that increased funding for quantum R&D "will be necessary for the nation to win the race to realize the benefits of QIS."[94] The Energy Sciences Coalition—"a group of more than 100 organizations representing scientists, engineers, and mathematicians from industry, universities, and national laboratories"—suggested increasing authorization funding levels for DOE alone to at least $675 million per year to maintain U.S. competitiveness in QIS.[95]

While the ideal level of funding is not clear, government is an important source of patient capital to provide long-term support for quantum technology and overcome high initial R&D costs. Government investment in QIS plays a critical role in developing a national network for research and design and facilitating the translation of research into real-world applications and technology commercialization.[96] It also plays an important role in conveying the potential and importance of quantum technologies to various stakeholders, including industry, the media, and the public. Federal spending levels are a strong indicator of government priorities, and U.S. government budgets for QIS R&D have remained strong and consistent for several years, with no signs of dramatically decreasing in the near term.[97]

Public funding and support for QIS has also played a role in expanding the number and type of institutions involved in quantum computing R&D. For example, NSF funds five Quantum Leap Challenge Institutes, which are "large-scale, university-led interdisciplinary research institutes to advance specific frontiers of quantum science and engineering."[98] Participating universities include the University of Maryland, Duke University, the University of Chicago, and the California Institute of Technology, among others. NSF also funds several university-directed quantum foundries, which are "specialized research centers developing and manufacturing the materials required for quantum technologies, such as qubits and sensors."[99] In August 2023, NSF also launched the $38 million Expanding Capacity in Quantum Information Science and Engineering (ExpandQISE) program to "lower barriers to access and broaden the diversity of institutions" participating in quantum R&D.[100]

Separately, DOE has helped engage national laboratories in quantum R&D efforts through the creation of five National Quantum Information Science Research Centers, which include:

- The Co-Design Center for Quantum Advantage (C2QA), led by Brookhaven National Laboratory, focused on building the tools necessary to create fault-tolerant quantum computer systems;
- The Next Generation Quantum Science and Engineering (Q-Next), led by Argonne National Laboratory, focused on how to reliably control, store, and transmit quantum information;
- The Quantum Science Center (QSC), led by Oak Ridge National Laboratory, focused on advancing quantum materials, sensors, and algorithms;

- The Quantum Systems Accelerator (QSA), led by Lawrence Berkeley National Laboratory, focused on developing scalable quantum systems; and
- The Superconducting Quantum Materials and Systems Center (SQMS Center), led by Fermi National Accelerator Laboratory, focused on developing superconducting materials and devices for next-generation quantum computers.[101]

DOE has also funded various quantum R&D collaborations between national laboratories, universities, and research institutes. In July 2023, for instance, DOE "announced $11.7 million in funding for six projects to improve understanding of whether, when, and how quantum computing might advance the frontiers of computational science."[102]

Universities, research institutes, and national laboratories are critical nodes in the U.S. quantum technology ecosystem and have played key roles not only in technology development, but in expanding QIS training, education, and outreach. However, these entities face significant administrative hurdles and onerous federal grant requirements that prevent them from fully contributing to quantum technology R&D initiatives. A 2022 study by the NSF's National Science Board found that cumbersome federal requirements on U.S. research universities impede the efficiency and quality of university science and engineering research.[103] The NQIAC further conveyed to Congress in March 2023 that a lack of designated professional administrative staff at NSF institutes required quantum-focused employees to assume administrative responsibilities, adversely impacting progress on funded quantum research projects.[104]

University professors and research institute directors interviewed for this project further noted that federal funding for quantum research often requires grant recipients to undertake educational outreach and engagement efforts that they are not trained to execute, and which distract time and resources away from the quantum R&D initiatives envisioned by the grant.[105] They also cited difficulty accessing federal resources for quantum R&D. For example, while DOE's National Research Centers aim to make quantum resources more accessible to a broader range of users, individuals interviewed for this report observed that traveling to such centers is cumbersome and inefficient. They suggested that a more effective approach may involve broadly distributing resources to different universities across the country, rather than concentrating resources in five national centers.[106]

U.S. Government QIS Ecosystem[107]

Coordination, Oversight, and Support

- U.S. House of Representatives and U.S. Senate
- National Quantum Coordination Office, Office of Science and Technology Policy
- Subcommittee on Quantum Information Science
- Subcommittee on the Economic and Security Implications of Quantum Science
- National Quantum Initiative Advisory Committee
- U.S. Department of State
- Federal Bureau of Investigation
- U.S. Patent and Trademark Office

Civilian R&D Entities

- National Science Foundation
- Department of Energy
- National Institute of Standards and Technology
- NASA

Military R&D Entities

- Defense Advanced Research Projects Agency
- Army Research Office
- Air Force Office of Scientific Research
- Office of Naval Research
- Army Research Laboratory
- Naval Research Laboratory
- Air Force Research Laboratory
- Office of the Under Secretary of Defense for Research & Engineering

Intelligence R&D Entities

- Intelligence Advanced Research Projects Activity, Office of the Director of National Intelligence
- Laboratory for Physical Sciences, National Security Agency
- Central Intelligence Agency
- Department of Homeland Security

Federally Funded National Laboratories

- Argonne National Laboratory
- Brookhaven National Laboratory
- Lawrence Berkeley National Laboratory
- Lawrence Livermore National Laboratory
- Fermi National Accelerator Laboratory
- Los Alamos National Laboratory
- Sandia National Laboratories
- Oak Ridge National Laboratory
- Pacific Northwest National Laboratory

PRIVATE SUPPORT FOR QUANTUM COMPUTING

The private sector has also played an increasingly prominent role in driving quantum technology innovation and the development of a robust quantum technology ecosystem in the United States. Private investment in quantum technology has steadily increased since 2013, hitting a record high globally of $2.2 billion in 2022.[108] However, venture capital (VC) investment dropped approximately 50 percent in 2023 to around $1.2 billion globally, prompting some concern from technology experts about a looming "quantum winter" of dwindling interest and investment in the technology.[109] But such concerns may be exaggerated and premature. The decline in quantum investments aligns with macro VC trends and financial downturns in other technology areas, such as biotechnology. Despite an overall decrease in VC investment, quantum technology witnessed 21 new disclosed private funding rounds in the second quarter of 2023, $329 million in new private capital in the third quarter of the year, and an increase in government-backed funding commitments and contracts, which could offset any gaps in investor apprehension.[110]

VC and other private capital constituted more than 70 percent of global quantum technology investments as of June 2022, with nearly 90 percent of this funding going to established Series A, B, C, and D start-ups.[111] U.S. companies, specifically those focused on quantum computing, received the greatest share of private funds between 2001 and 2021, and five of the eight top-funded quantum technology companies in 2022 were headquartered in the United States.[112] Approximately 49 percent of private funds in 2022 went toward the quantum computing component area.[113] Heightened interest in quantum computing relative to other quantum technology areas is likely due to quantum computing's tremendous market potential, which is estimated to be upward of $90 billion annually by 2040.[114] It is important to note, however, that despite a significant increase in activity in recent years, quantum technology still represents less than 1 percent of global VC investment.[115] Several investors interviewed for this report noted that venture capitalists' reluctance to enter the quantum industry typically stems from uncertainty about anticipated technology development timelines, returns on investment, and which applications will ultimately prove profitable.[116]

In 2023, venture funding for quantum in the United States declined approximately 80 percent from 2022 levels, marking the largest decrease globally.[117] As previously noted, this decline aligns with broader trends in the venture market. Further, 2021 and 2022 were notable years of significant, perhaps unsustainable quantum technology investments with large special-purpose acquisition companies and funding rounds. The decline in funding in 2023 could thus represent recalibration back to pre-2021 figures. According to The Quantum Insider, a quantum technology media and market intelligence platform, the future outlook for the quantum industry is positive, suggesting "a continued interest in quantum technologies" with "expectations of more technical and engineering breakthroughs, increased government procurement, and a possible evolution in venture capital focus towards deep tech."[118]

Investors' relatively consistent interest and investment in quantum technology has sparked significant growth in the U.S. quantum industry. The NQI-funded Quantum Economic Development Consortium (QED-C)—a group of stakeholders "that aims to enable and grow the quantum industry"—contained 182 corporate members as of late 2023.[119] Prior studies determined that nearly half of QED-C companies work on quantum computing and that the United States is the most active country globally in quantum computing, as measured by the number of entities involved in quantum computing R&D.[120]

Though most U.S. quantum computing companies are component and application software companies, hardware start-ups receive the largest share of funding.[121] This is likely because hardware is capital intensive; it is complex, requires specialized expertise, entails significant risk, and demands long development timelines. At the same time, investors anticipate that hardware will generate significant returns in the application and commercialization phase of quantum technology development.[122] Large technology players like IBM dominate the hardware investment landscape and primarily focus on superconducting qubits.[123] However, the U.S. quantum computing industry is broad—the United States is home to at least one company pursuing each quantum computing modality.[124] This diversified approach helps ensure that the United States has the technical knowledge, domestic infrastructure, and R&D capacity to be at the leading edge of whichever modalities win the race to build the first commercially viable quantum system. Other countries similarly pursue a variety of modalities simultaneously, but no one country has a level of breadth and depth comparable to that of the United States.

China's Quantum Technology Strategy

China's objectives in quantum technology are intertwined with its broader ambition to become a "global leader in innovation."[125] Chinese President Xi Jinping

views technology as a means to enhance China's economic and military capabilities, bolster national power, and emerge as the global "science and technology power." Xi views progress in quantum technology as key to fulfilling this vision.

Although the Chinese Communist Party (CCP) has long supported basic quantum research, its interest in quantum intensified in 2013 after former U.S. National Security Agency contractor Edward Snowden leaked classified documents detailing the United States' intelligence capabilities and activities in China. As Elsa Kania and John Costello noted in a 2017 report, Snowden's revelations sparked "anxieties over domestic information security and vulnerabilities to cyber espionage and influence, provoking a search for new cyber security solutions developed indigenously."[126] Chinese leaders subsequently established the development of quantum technologies as a national priority, noting their potential to enable the "absolute security" of sensitive information and communications and "reset the military and intelligence balance in China's favor."[127]

China lacks a public national strategy for quantum similar to the United States' National Quantum Strategy. Instead, China promotes quantum technology R&D through a series of broader national science and technology plans and projects. The 13th Five-Year Science and Technology Military-Civil Fusion Special Projects Plan, for example, highlighted the importance of quantum technology and launched a "megaproject" for quantum communications and computing focused on "driving major quantum technology breakthroughs," including the "expansion of China's national quantum communications infrastructure, the development of a general quantum computer prototype, and the construction of a practical quantum simulator."[128] The 13th Five-Year National Science and Technology Innovation Plan similarly prioritized advancements in quantum communications and computing, including the development of a universal quantum computing prototype and practical quantum simulator.[129] All of China's quantum-relevant plans and projects reflect the CCP's intent to achieve key technology breakthroughs by 2030 and apply quantum technologies to defense and security challenges.

Public comments from high-level government officials reinforce these documents and underscore China's primary quantum technology objectives: self-reliance and global leadership.[130] In a May 2024 interview with the *South China Morning Post*, Guo Guoping—a deputy of China's National People's Congress and founder of Chinese quantum company Origin Quantum—stated

Chinese President Xi Jinping views technology as a means to enhance China's economic and military capabilities, bolster national power, and emerge as the global 'science and technology power.' Xi views progress in quantum technology as key to fulfilling this vision.

that "building an independent industry chain" for the quantum sector and overcoming the "increasing technological containment of Western nations" have become "China's top priorities."[131]

Further, in remarks before the Political Bureau of the CCP Central Committee in October 2020, Xi referred to quantum as a "major disruptive technological innovation that impacts and restructures the traditional technological system" and will "lead a new round of sci-tech revolution and industrial transformation." He noted China's intent to "gain an upper hand in international competition" and "build new advantages" in quantum while stressing "the importance of strengthening strategic planning and systematic layout for the development of quantum science and technology." Xi noted that China aims to "follow the path of independent innovation, make breakthroughs in key core technologies, and ensure the safety of industrial and supply chains." He also highlighted the importance of talent, calling for the development of quantum training programs and curricula.[132] In line with these objectives, the CCP has made significant investments to bolster the country's quantum technology ecosystem.

PUBLIC SUPPORT FOR QUANTUM COMPUTING

Authoritative public information on total levels of government funding for quantum R&D in China is limited, and different sources provide drastically different estimates.[133] A June 2022 McKinsey & Company report estimated that China has earmarked a total of $15.3 billion for quantum technology, placing it well ahead of the European Union's $7.2 billion and the United States' $1.9 billion in planned public funding.[134] A 2022 Wilson Center report similarly estimated that China has devoted over $15 billion to quantum technology, but noted that "doubts persist on the accuracy of these amounts."[135] A 2023 analysis by The Quantum Insider also valued

China's state investment at $15 billion. If accurate, this means that China's state funding comprises more than 50 percent of the world's public spending on quantum.[136]

Other estimates attempt to capture China's year-to-year investments yet similarly fail to produce consensus. The Congressional Research Service (CRS) estimated China's annual funding for quantum technology R&D at $244 million.[137] As observed in a recent RAND Corporation study, however, quantum experts within China report much different figures. In a November 2019 academic article, Jian-Wei Pan—a prominent quantum technology researcher who is often referred to as China's "father of quantum"—reported that quantum technology R&D expenditures in China were significantly lower than the CRS's estimate, totaling $84 million between 2016 and 2020 (as of November 2019), $98 million between 2011 and 2015, $30 million between 2006 and 2010, and $1.25 million between 1998 and 2006.[138]

At the same time, Chinese media outlets have suggested that expenditures are higher than the CRS's estimate. The Anhui Business Daily Newspaper reported that the CCP planned to spend $2.95 billion annually on quantum between 2017 and 2022, while other Chinese-language news media reported $1.06 billion in state funding for a single Chinese quantum research facility—the Hefei National Laboratory for Physical Sciences at the Microscale, where most Chinese quantum computing research occurs—in 2017 alone.[139] In 2017, the Chinese government also directed the construction of the $10 billion, 91-acre National Laboratory for Quantum Information Sciences, which is now the world's largest national laboratory for quantum science.[140]

In short, official reports of Chinese state funding for quantum R&D vary dramatically. According to RAND, "these conflicting reports of funding levels are not unusual in China." The CCP "often announces ambitious (and often highly politicized) spending goals, and it is not uncommon for these goals to go unmet."[141] Although exact figures are uncertain, most analysts agree that China is outspending the United States on quantum technology.[142] In April 2023, The Quantum Insider concluded that "China has significantly higher public spending" compared to the United States, spending approximately four times more.[143] Previous studies observe that like in the United States, government QIS R&D investment in China appears to be relatively stable and consistent. Though China lacks public legislation similar to the United States' NQI that "formally authorizes multiyear appropriations for specific purposes," China's five-year plans and related planning documents "indicate that quantum technology has been a sustained strategic priority" of the Chinese leadership that is only "increasing in importance."[144]

Various Chinese government agencies and state-linked organizations are involved in driving QIS R&D, including the Ministry of Science and Technology, the Ministry of Industry and Information Technology, the Chinese Academy of Sciences, and the China Construction Bank.[145] State-linked defense entities—such the Aviation Industry Corporation of China and the Academy of Military Science—have also engaged in quantum R&D efforts, and quantum technology is included in China's broader national strategy for civil-military fusion. The 13th Five-Year Science and Technology Military-Civil Fusion Special Projects Plan, for example, highlighted quantum computing and communications projects. But as RAND physical scientist Dr. Edward Parker noted in recent congressional testimony, China's "overall priorities, plans, and timelines for integrating quantum technology into operational military systems are much less clear than those of the U.S. government."[146]

China has established several collaborative institutes and projects to explore military applications of quantum technologies, including the Front-Line Cross-Disciplinary Technologies Research Center, the Beijing Academy of Information Science, and joint University of Science and Technology of China–China Shipbuilding Industry Corporation quantum laboratories.[147] But quantum radar—which the U.S. DoD considers "scientifically impractical"—is one of the only military-specific quantum technologies that China is explicitly pursuing.[148] Though China reportedly has a successful quantum radar prototype, there are several challenges associated with its successful implementation. It is not clear, for example, whether China will be able to integrate quantum radars into existing platforms, or whether quantum radars will provide enough of an advantage to justify China's investments.[149]

Further, China's military-affiliated universities—including the People Liberation Army (PLA) Academy of Military Sciences and PLA National Defense University—"are not at the forefront of Chinese public research in quantum science."[150] Only one military-affiliated university—The National University of Defense Technology—is included on the list of China's top 20 publishers of quantum technology research.[151] While the CCP certainly appreciates the role that quantum technology could play in enhancing its military power and will continue exploring military applications of quantum, on balance, China appears to prioritize intelligence and civilian applications at this stage of development.

Like in the United States, public funding for QIS R&D in China is spread across multiple quantum technology component areas. Driven by its desire to improve information security in the wake of the Snowden leaks, China has prioritized funding for quantum communications and is considered the global leader in the development and deployment of quantum communications technologies. In recent years, however, China has increased the share of funds it devotes to the quantum computing component area as part of a broader push to assume leadership in this realm.

The Quantum Insider estimates that China now devotes as much as 26 percent of QIS R&D funding to quantum computing and 22 percent to hardware development.[152] Research from RAND further demonstrates that while quantum communications still maintains the widest breadth of state investment sources in China with 27 distinct funding sources, quantum computing is a close second with 24 distinct sources.[153] The level of quantum computing research activity in China matches this funding trend. Between 2010 and 2019, the number of quantum computing patent applications filed in China grew 40 percent, while quantum communications and quantum sensing patents grew 38 and 29 percent, respectively.[154]

Academic journal publications, reporting in local media outlets, and previous studies indicate that the "bulk of Chinese research on quantum computing is done at the Hefei National Laboratory for Physical Sciences at Microscale."[155] Researchers and engineers in China appear to focus on the four most common technical approaches to quantum computing, which were previously outlined in this report—superconducting, photonics, neutral atoms, and trapped ions.[156] According to several technical experts, China's development and deployment of superconducting qubits is "at rough parity with the United States."[157] China is also considered the "clear world leader in photonic boson sampling," a simplified approach to linear optical quantum computing. Though boson sampling cannot produce a full-scale, fault-tolerant quantum computer, it could be a useful tool for experimentation and education.[158]

China's expansive approach to hardware R&D has yielded results across multiple modalities. China reportedly possesses two quantum computers—the photonics-based Jiuzhang 2 computer and superconducting-based Zuchongzhi 2 computer—that outperform Google's Sycamore quantum computer, a 54-qubit transmon superconducting processor capable of performing a calculation in 200 seconds that would take a classical computer thousands of years to complete.[159] The Jiuzhang 2's computational speed and power allegedly exceeds the Sycamore computer "by factors thought to be in the billions," while the Zuchongzhi 2 "cracked a problem three times tougher" than what the Sycamore could solve.[160]

In 2021, China also became the third country—behind Canada and the United States—to develop and deliver a completely domestically built superconducting quantum computer.[161] This milestone illuminates China's broader aim to develop indigenous development capabilities and ensure technological independence. In addition to prototype quantum computers, Chinese researchers are pursuing algorithmic development. In December 2022, Chinese researchers announced that they had designed an algorithm capable of breaking public-key encryption, years before anticipated.[162] Many experts question the validity of the announcement, but nonetheless, it serves as a reminder that "China could achieve surprising quantum computing breakthroughs in the near term."[163]

China has also reached milestones in other quantum technology component areas that could support and expedite its development of quantum computing. For example, the Chinese Academy of Sciences launched its Quantum Experiments at Space Scale Micius satellite—the world's first quantum satellite—in 2016. The Micius launch enabled China to deploy quantum key distribution (QKD), a type of quantum-based encryption that allows for secure communications over long distances. China has used Micius for a range of experiments and now maintains a quantum network that includes two quantum satellites and approximately 700 fiber-optic cables.[164]

The U.S. National Security Agency and several quantum experts question the value of QKD, highlighting the technology's various technical limitations.[165] But China's execution of QKD at scale demonstrates its ability to overcome challenging technological requirements. A January 2024 report by the Mitchell Institute noted that China's progress in QKD indicates that it can "produce and encode on-demand photonic qubits, maintain signal strengthen across thousands of miles using both free-space and fiber-optic transmission means, and develop secure and accurate repeater stations."[166] This knowledge and expertise could potentially give Chinese researchers and engineers an advantage when tackling analogous challenges associated with quantum computing. China's existing capability in quantum communications could also help it deploy other quantum technologies faster. As the Mitchell Institute observed, China's quantum communications network "serves as a foundational infrastructure to discover,

understand, and solve the problems that become clear only once a capability moves out of the laboratory and into the real world." The United States "does not currently have a similar architecture."[167]

PRIVATE SUPPORT FOR QUANTUM COMPUTING

Private industry assumes a less prominent role in China's quantum technology ecosystem than U.S. industry assumes in the United States' ecosystem. Nonetheless, Chinese research in quantum computing "has started to involve a higher degree of private sector involvement and investment, motivated by the tremendous commercial potential of quantum computers."[168] A sample list of the entites involved in quantum technology R&D in China can be found on page 20.

Just as it is difficult to determine the level of state funding for QIS R&D in China, information about which Chinese firms are meaningful participants in China's quantum activities is opaque and contradictory. According to a Chinese business report, more than 4,200 firms claimed to work on quantum technology as of October 2020.[169] Further examination by RAND, however, revealed that most of these firms are "'quantum' in name only." After evaluating Chinese media, quantum industry publications and media reports, and patent searches, RAND identified only 13 Chinese firms engaged in "meaningful R&D in quantum technology," including five large, diversified technology companies and eight specialized quantum technology start-ups. Three of the start-ups focus exclusively on quantum computing, while another three focus on quantum communications, one focuses on quantum sensing, and one tackles multiple component areas. Though Chinese quantum technology start-ups are similar in age to U.S. start-ups, Chinese companies have announced significantly lower levels of capital funding than their U.S. counterparts.[170]

A study by the QED-C similarly concluded that China's quantum industry is smaller than Chinese business reporting suggests. The QED-C identified only nine Chinese companies that "have material quantum computing R&D capabilities."[171] The nine identified companies specialize in various quantum computing modalities, including superconducting, photonics, trapped ions, and neutral atoms. This diversity signals China's desire to reach the cutting edge of quantum R&D and develop indigenous capabilities across the spectrum of quantum computing hardware. Pursuing multiple quantum computing modalities could also bolster the Chinese quantum industry's resilience to export controls—because each modality requires different components and materials, it will be difficult for China's adversaries to identify chokepoint controls that impact the country's entire quantum computing ecosystem.

Though the private investment environment remains significantly smaller than state spending, China's private investments in quantum are increasing. According to The Quantum Insider, investment in Chinese quantum start-ups reached $194 million in 2022, a marked increase from the $62 million invested in 2021. The majority of investments in 2022—approximately 66 percent—were devoted to companies focused on superconducting quantum computers.[172] Origin Quantum Computing had a particularly notable year, raising $148 million in a Series B funding round.[173] Several government-affiliated investors led by the China Internet Investment fund contributed to the funding round.[174]

At the same time, several of China's largest technology companies appear to be losing interest in quantum. Chinese company Baidu, one of the largest internet companies in the world, announced the termination of its quantum R&D efforts in January 2024. Baidu—which was primarily focused on the development of quantum computing software and hardware solutions, including a quantum hardware-software integration system—reportedly plans to donate its quantum computing laboratory and equipment to the state-linked Beijing Academy of Quantum Information Sciences.[175] Two months prior, Alibaba, another Chinese technology giant, cut its quantum computing research portfolio—which was primarily focused on open-source initiatives and quantum algorithm development—and donated its laboratory and related experimental equipment to the state-linked Zhejiang University.[176]

Neither Baidu nor Alibaba provided a public explanation for the termination of their programs, but there are several factors that could have influenced their decisions. The two companies donated their quantum computing laboratories and equipment to state-linked entities with robust government-funded quantum R&D programs. It is thus plausible that the CCP compelled Baidu and Alibaba to relinquish their quantum capabilities as part of a broader government effort to centralize and gain more control over quantum R&D activities. Alternatively, Baidu and Alibaba may seek to focus on developing more advanced emerging technologies with distinct business applications, such as artificial intelligence, rather than quantum, which is expensive and resource intensive with no clear use cases or immediate return on investment.

Another possibility is that Baidu and Alibaba want to avoid developing clearly demonstrated progress in domestic quantum computing due to increasingly strong U.S.-based export controls in areas related to advanced

computing, including semiconductor manufacturing equipment, processor design software, and accelerators.[177] The United States has also already implemented some quantum-specific export controls targeting certain quantum sensing capabilities and China-based end users, and U.S. officials have signaled that additional quantum technology controls may be forthcoming.[178] Visible demonstrations of world-class quantum computing R&D capabilities that threaten U.S. dominance in the field could thus subject Baidu and Alibaba to strict U.S. policies that limit the flow of key quantum computing-related intellectual property, materials, or subcomponents.

Regardless of the motivation behind the decisions, Baidu and Alibaba were two significant players in China's already small private quantum technology industry and their departure from the quantum R&D landscape could adversely affect China's overall advancement in quantum technology. Large, diverse companies have unique understandings of technology markets and business requirements and will thus likely play important roles in the quantum commercialization process.

Takeaways

While the United States and China share the goal of achieving global leadership in quantum technology, their approaches to quantum technology R&D differ dramatically. The United States takes an industry-led, government-supported approach. The U.S. quantum industry, which includes a mix of established technology companies and recently founded start-ups, tends to drive innovation and key development milestones. In China, quantum technology R&D is generally government-led and industry-supported. China's quantum industry is significantly smaller than the United States', and most quantum breakthroughs in China stem from government-funded laboratories and research institutes.

The United States and China both pursue a variety of quantum computing modalities, including superconducting, trapped ions, photonics, and neutral atoms. According to several technical metrics—including quantum computers' physical qubit counts, coherence times, and error rates—the United States leads in every modality except superconducting transmon qubits, an area in which China is at rough parity with the United States.[180] The United States has developed high-performing prototypes across all modalities and is exploring a wide range of applications, from scientific discovery and materials research to logistics and advanced electronics. China's technical progress is largely confined to superconducting qubits and photonic boson sampling,

China's QIS Ecosystem[179]

Government Bodies and National Laboratories

- Ministry of Science and Technology
- Ministry of Industry and Information Technology
- National Development and Reform Commission
- National Natural Science Foundation of China
- Industrial and Commercial Bank of China
- China Construction Bank
- China Internet Investment Fund
- China Reform Fund
- Municipal governments of Shanghai and Beijing
- Hefei National Laboratory for Physical Sciences at the Microscale
- National Laboratory for Quantum Information Sciences

Universities and Research Institutes

- Chinese Academy of Sciences
- Tsinghua University
- University of Science and Technology of China
- Beijing University of Posts and Telecommunications
- Nanjing University
- Peking University
- Shanghai Jiao Tong University
- Zhejiang University
- National University of Defense Technology
- Shanxi University
- Dalian University of Technology
- Anhui University
- Front-Line Cross-Disciplinary Technologies Research Center

Companies

- Huawei Technologies
- CIQTEK
- Origin Quantum
- Qasky
- QuantumCTek
- QuDoor (also referred to as Guokaike Quantum Technology and Qike Quantum)
- Tencent Quantum Lab
- ZTE Corporation
- Kunfeng Quantum Technology
- SpinQ
- Shenzhou Quantum Communication Technology
- Bose Quantum
- TurningQ

a nonuniversal approach to quantum computing. Chinese researchers are particularly focused on developing encryption-breaking applications.[181]

Though total R&D expenditures, patents, and research publications are imperfect and potentially misleading measures of a country's capacity for technology development and diffusion, they can help further illuminate the United States' and China's potential in quantum computing.[182] By most estimates, the Chinese government dramatically outspends the U.S. government on quantum technology R&D, but the United States maintains a higher level of private investment. The United States is also the global leader in quantum computing patent applications according to a February 2023 QED-C study. The United States Patent Office received approximately 1,838 quantum computing patent applications between 2010 and 2022, while China's National Intellectual Property Administration received only 908.[183] During the same timeframe, the United States also had the largest number of entities among the top 19 entities that filed quantum computing patents. Of the top 19 patentees, nine were based in the United States and five were based in China.

U.S. entities similarly dominate the list of the top 20 entities filing international quantum computing patents. International patents, which are filed in patent offices across multiple countries, are more difficult to acquire and typically reserved for higher-quality patents.[184] As of 2021, IBM had the largest number of international patent families—401. Nine additional U.S. entities were included on the top 20 list of international patentees. Only one Chinese company—Alibaba—made the list, and Alibaba reportedly terminated its quantum computing R&D efforts in November 2023.[185]

The volume and quality of research publications are other metrics to assess a country's potential to achieve key breakthroughs in quantum computing. China produces the highest volume of quantum computing research, accounting for approximately 22.8 percent of research globally, while the United States ranks second, accounting for approximately 21.3 percent of global research. However, the Hirsch index (H-index), which measures total publications and the number of citations a publication receives, indicates that the United States produces higher quality, more impactful research than China. The United States leads the world in quantum research quality with an H-index of 92. China ranks second with an H-index of 52.[186]

Taken together, these metrics suggest that the United States is the global leader in quantum computing but faces stiff competition from China. Though China lags behind the United States in overall technical progress, Chinese experts produce a high volume of quantum-relevant research and receive a high level of government support. China clearly intends to surpass the United States in quantum computing and is rapidly assuming the capabilities to do so.

It is not too late for the United States to secure its lead in quantum computing, but policymakers must act quickly to maximize U.S. strengths and patch existing vulnerabilities. To ensure American competitiveness in quantum computing, U.S. policymakers should focus on supporting the development of a secure and resilient quantum technology supply chain and building a diverse quantum workforce.

Though China lags behind the United States in overall technical progress, Chinese experts produce a high volume of quantum-relevant research and receive a high level of government support.

Securing U.S. Leadership in Quantum Computing

Though U.S.-China competition in quantum computing raises several important policy questions—including questions about export controls, outbound investment restrictions, and technology standards—the United States cannot effectively address these areas without first tackling supply chain and workforce challenges. Absent a secure and resilient quantum supply chain and a well-trained workforce, the United States lacks the leverage, technical knowledge, and scientific edge required to craft and implement effective policies in other areas. In the near term, the United States can ensure competitiveness in quantum computing by conducting a comprehensive supply chain review, leveraging financial incentives to boost domestic production of sensitive quantum technology inputs, sharing resources with allies and partners, creating new educational opportunities, and fostering public-private cooperation.

Building a Robust Quantum Technology Supply Chain

A well-developed quantum technology supply chain with limited dependence on foreign suppliers is critical to U.S. leadership and competitiveness in quantum technology. As President Biden's Executive Order on America's Supply Chains notes, "resilient, diverse, and secure supply chains" are "fundamental to U.S. economic prosperity and national security."[187] A robust quantum technology supply chain will boost American manufacturing capacity, help maintain the United States' competitive edge in R&D, and create high-paying jobs. It will also help insulate the United States from geopolitical and economic competition, cyberattacks, and other potential disruptions.[188]

To foster a resilient quantum computing supply chain, policymakers must first understand which components and materials are essential and how the supply chain is currently structured. They must also identify vulnerabilities and assess the likelihood of potential disruptions. The following section offers an overview of the quantum computing supply chain and highlights the U.S. quantum ecosystem's primary supply chain challenges.

One key issue for the quantum supply chain is a lack of clarity about which raw materials, components, manufacturing equipment, and supporting infrastructure are required to make quantum systems function and therefore warrant U.S. government investment and protection. As previously outlined in this report, the quantum computing sector is nascent, with several different modalities under development. Each modality requires a different and evolving set of inputs, with very little overlap. While superconducting qubits require helium dilution refrigerators, other modalities, like trapped-ion qubits, instead rely on high-quality lasers and isotopically pure samples of various elements. This means that dilution refrigerators could be critically essential or completely irrelevant depending on which quantum computing modalities prevail. Unless the United States is prepared to produce every input for every modality, an entirely domestic U.S. supply chain is unrealistic.[189]

Further, the inputs on which quantum technologies depend could change as quantum technologies mature. Researchers and engineers can speculate about which components, materials, and subsidiary technologies will be required during the next phase of quantum computing development, but a comprehensive list of enabling equipment is impossible to compile until full-scale quantum computers are readily available and integrated into existing infrastructure. The establishment of a "well-defined, complete, and stable" quantum technology supply chain will thus "remain in flux" for the next several years or even decades.[190]

Though there are still many unknowns, a 2022 Hyperion Research survey of 47 quantum technology companies provides some insight into which materials, components, or subassemblies are most important and could present supply chain challenges in the near term. The Quantum Computing table below outlines the products respondents identified as the most likely potential causes of a quantum computing supply chain disruption in the next three years.[191]

Quantum Computing Materials, Components, and Subassemblies[192]

Materials	Components and Subassemblies
▪ Helium-3 gas ▪ Silicon-28 ▪ Copper ▪ Aluminum ▪ Gold	▪ High-performance cryocoolers ▪ Pumps ▪ Valves ▪ Compressors ▪ Power supplies ▪ RF generators ▪ Superconducting wiring assemblies ▪ Dilution fridge components ▪ Fiber and coaxial cables ▪ Low-noise lasers ▪ Key manufacturing equipment for quantum and classical chip manufacturing and testing

Another challenge is that quantum technology supply chains are global and highly specialized, with many single points of failure.[193] Hyperion Research's 2022 survey revealed that a majority of quantum computing companies lack supply chain diversity. Companies often have only one supplier of their most critical manufacturing components, and finding an alternative supplier would take more than a year.[194] Additionally, many critical suppliers reside outside of the United States. One U.S. quantum company interviewed for this report indicated that 80 percent of its supply chain falls outside of the United States.[195] The United States maintains several foreign dependencies, which previous studies attributed to "relative component costs, differences in quality, a lack of domestic alternatives, and acquisition of domestic suppliers by foreign companies."[196]

Countries in Europe and Asia are developing outsized influence over the availability of components needed to build and sustain quantum computers, including cryostats, synthetic diamonds, and various lithography tools. This means that continued U.S. progress in quantum computing increasingly depends on the health and stability of a few key foreign suppliers. Should any one of these suppliers experience an operational disruption or decide to limit or terminate sales to the United States, it could take the United States years to find or develop alternative options and significantly handicap U.S. quantum R&D efforts.

Dilution refrigerators, for example, are part of a very small industry located primarily in Europe. Finland-based Bluefors Oy, United Kingdom–based Oxford Instruments, and Netherlands-based Leiden Cryogenics BV are currently the leading producers of dilution refrigerators used for quantum computing.[197] A handful of U.S. companies—including California-based Quantum Design and Colorado-based Maybell Quantum—are developing the capabilities to produce cryogenic systems. However, several quantum technology experts interviewed for this report—including university professors and quantum startup CEOs—noted that these companies are relatively immature and require additional financial support to succeed.[198] China similarly appreciates the supply chain challenges associated with dilution refrigerators and is developing its own independent capabilities in an effort to reduce "reliance on international manufacturers" and "lead in the quantum computing field."[199] In early 2024, a leading Chinese news service reported that China "has begun mass production of the 'EZ-Q Fridge,' a dilution refrigerator essential for superconducting quantum computer chips."[200]

Europe serves as a hub for several other important quantum computing inputs as well. The United Kingdom is a key source of specialized synthetic diamonds and monolithic integrated windows for vacuum chambers.[201] Germany is an important source of high-quality lasers, dielectric glass windows, and single-photon detectors. France produces fiber phase modulators and double-angle evaporators. Sweden provides high-electron mobility transistor amplifiers, and the Netherlands is a leading producer of advanced lithography equipment. Russia is the leading producer of large sapphire wafers used in some trapped-ion quantum computing systems, as well as in the manufacturing of superconducting transmon quantum systems.[202] In essence, the United States relies on suppliers across Europe for inputs that are necessary to all four leading quantum computing modalities—superconducting, trapped ions, neutral atoms, and photonics.

Asia is another critical supplier region. Japan is a source of photon detectors, blue gallium nitride diodes, nonlinear optics, and electron beam lithography tools.[203] China, the Philippines, Taiwan, and Malaysia produce microcontrollers. China is also a source of less-advanced electronics and optics, as well as rare-earth elements, which are input to many quantum technology components.[204] The magnet materials in the compressors used to precool dilution refrigerators are composed of rare-earth elements, for example. China accounts for "63 percent of the world's rare-earth mining, 85 percent of rare-earth processing, and 92 percent of rare-earth magnet production."[205]

The quantum supply chain is nimble. Policymakers and industry stakeholders still have an excellent opportunity to shape the supply chain in a way that favors U.S. interests.

China has already demonstrated a willingness to use its advantageous position in the minerals supply chain as geopolitical leverage.[206] As recently as 2023, Chinese officials reportedly contemplated banning the export of certain rare-earth processing and refining technologies to the United States.[207] A loss of access to rare-earth elements and other critical minerals would carry implications beyond the quantum sector—without critical minerals, U.S. innovation would stall, the cost of consumer electronics would surge, and the U.S. military's weapons would fail to hit their targets.[208] The rare-earth challenge thus highlights the risk associated with depending on competitors for critical components.

The quantum technology supply chain is highly dispersed and specialized with several single points of failure, presenting significant difficulties for the U.S. quantum industry and U.S. policymakers. At the same time, however, the quantum supply chain is nimble. Policymakers and industry stakeholders still have an excellent opportunity to shape the supply chain in a way that favors U.S. interests. To foster the development of a secure and reliable supply chain, the U.S. government should:

Conduct a comprehensive review of the quantum technology supply chain.

The recently established White House Council on Supply Chain Resilience (the Council) should initiate an in-depth survey of the current quantum technology supply chain to identify critical nodes and assess the United States' level of dependence on foreign suppliers. The Council should compile its findings into a report that details the components, materials, and subassemblies required for each quantum computing modality, identifies the suppliers of those inputs, and assesses suppliers' capacity for consistent, reliable production and future growth. The report should also identify opportunities to develop additional suppliers across a range of regions, as well as the United States' potential to build domestic production capacity for key inputs. The Council should consult members of the QED-C, quantum research institutes, and academic researchers and engineers to gain a bigger picture of the U.S. quantum ecosystem's supply chain challenges and concerns.

This information will help policymakers manage and mitigate risk. Because a completely domestic quantum technology supply chain is infeasible, the quantum technology supply chain is inherently risky—some degree of dependence on foreign suppliers is necessary. A comprehensive review of current and anticipated trends in the quantum supply chain will allow policymakers to choose where to accept risk and help shape the development of a resilient and secure supply chain that promotes U.S. interests. By illuminating potential chokepoints and vulnerabilities, it will also help U.S. policymakers design more effective export control regimes should they eventually decide that trade restrictions are necessary to slow or degrade China's quantum technology development.[209]

Leverage financial incentives and the Tech Hubs Program to promote domestic production of quantum technology inputs deemed too sensitive to reside outside of the United States.

The White House should leverage the Defense Production Act (DPA), as well as the Small Business Innovation Research (SBIR), Small Business Technology Transfer (STTR), and Tech Hubs programs to boost domestic capacity for the production of quantum computing components, materials, and subassemblies that are deemed too sensitive to reside predominantly outside of the United States.[210]

The Biden administration has already used the DPA to strengthen the nation's critical minerals supply chain. In February 2023, President Biden invoked the DPA to fund projects to increase the domestic availability of lithium, nickel, cobalt, graphite, and manganese to support the production of large-capacity batteries and enable the energy transition.[211] Former president Donald Trump similarly leveraged the DPA to improve domestic mining and refining capacity and increase production of samarium cobalt magnets.[212] The White House could model this approach to advance U.S. production of other quantum technology materials and components as well, such as helium or synthetic diamonds.

The White House can also use SBIR and STTR programs to increase funding for U.S. companies developing key quantum technology inputs and incentivize them to align their R&D efforts with U.S. government priorities.[213] The SBIR and STTR programs support could, for example, give emergent cryogenics companies a financial boost and help connect them with the broader U.S. quantum computing ecosystem.

The Tech Hubs Program—a CHIPS Act–authorized "economic development initiative designed to drive regional innovation and job creation by strengthening a region's capacity to manufacture, commercialize, and deploy technology that will advance American competitiveness"—presents another opportunity to strengthen and secure the United States' quantum technology supply chain.[214] In October 2023, the Biden administration announced two quantum Tech Hub Designees—Elevate Quantum, which "aims to solidify Colorado's global leadership in quantum information technology to increase infrastructure resilience and strengthen the quantum hardware supply chain," and the Chicago Quantum Exchange, which aims to advance a similar mission in Illinois, Indiana, and Wisconsin.[215] The White House should establish a group of independent external research teams to develop mechanisms

to objectively gauge the success and impact of these programs, and determine whether they can be replicated and scaled in other regions. The White House should consider establishing future quantum tech hubs in states with a growing number of quantum computing companies, such as Massachusetts and California.

Direct and resource the QED-C to continually monitor the U.S. quantum computing supply chain and foreign competitors' efforts to develop their own domestic quantum computing ecosystem.

While the Council's initial quantum supply chain survey will help assess near-term challenges, the United States also needs a long-term mechanism to detect new supply chain vulnerabilities as quantum technologies evolve. Early detection of emerging vulnerabilities will give policymakers a window of opportunity to intervene before the United States develops unsustainable dependencies on foreign suppliers.

Information about competitors' efforts to boost their own domestic infrastructure is also critical to supply chain security and American competitiveness in quantum computing. Understanding competitors' supply chain capabilities and intentions can help illuminate which quantum computing modalities they are pursuing, as well as opportunities for potential collaboration. It can also help U.S. policymakers ensure that foreign efforts to develop independent quantum computing ecosystems do not adversely affect the U.S. quantum industry's ability to fairly compete abroad.[216]

The QED-C unveiled a framework for a "practical and extensible supply chain tracking tool" in September 2023 and is thus well-positioned to assume responsibility for continuous supply chain monitoring.[217] The White House should resource the QED-C to build an initial proof of concept, which will allow the QED-C to refine the tool's functionality, determine whether the tool captures appropriate data sources, and develop efficient processes for collecting, verifying, aggregating, and updating data over time. The QED-C should partner with The Quantum Insider or similar intelligence platforms to develop a comparable framework for monitoring competitors' quantum supply chain efforts.

Establish an international forum to coordinate supply chains with U.S. allies and partners.

No single country has all the components and materials required to build a quantum computer. As such, countries must maximize their comparative advantages and cooperate to build a supply chain that is secure and reliable. The United States can adopt a global leadership position in quantum technology by creating a forum that helps identify opportunities for coordination and facilitate supply chain burden sharing.

The Quad Center of Excellence in Quantum Information Sciences (Center of Excellence) provides an excellent foundation from which to build an international supply chain forum. The Center of Excellence was established in June 2023 to bring "together subject matter experts, capital allocators, and business leaders" across the Quad countries—the United States, Australia, India, and Japan—to "drive greater technological collaboration, market access, and cross-border investments."[218] The U.S. NQCO should propose the establishment of a supply chain task force within the Center of Excellence to discuss how the Quad countries can work together to mitigate supply chain vulnerabilities and collectively increase production of critical components and materials. The OSTP and the Department of State should also host an international roundtable on quantum computing supply chain security similar to a dialogue the agencies held in May 2022 on international cooperation in quantum technology R&D.[219] The roundtable should gather the heads of quantum strategy offices around the world, as well as industry representatives and academic stakeholders.

Fostering a Quantum Technology Workforce

U.S. policymakers must also prioritize workforce development to secure and advance American competitiveness in quantum computing. The U.S. National Quantum Strategy notes that "growing an American quantum-smart workforce with expertise in a broad range of physical, information, and engineering sciences is crucial for assuring sustained progress in QIS."[220] Access to talent is required not only to invent, develop, and scale quantum technologies, but to identify quantum applications and ensure responsible technology use. A well-trained quantum workforce is "key to realizing the full practical value of quantum computing."[221] Access to talent will be a critical factor in determining which country prevails in quantum technology competition.

Because quantum technologies are still in an early stage of development and the industry's future trajectory is highly uncertain, the precise definitions of "quantum talent" and a "quantum workforce" are still unfolding. No authoritative data source for determining quantum workforce requirements exists, and the desirable jobs, skills, and degrees are ambiguous. It is increasingly clear, however, that a diverse talent pool representing various

disciplines and backgrounds is most conducive to long-term competitiveness in quantum technology. This includes both quantum-specific expertise—which is required to advance the basic science—and general science, technology, engineering, and mathematics (STEM) talent—which is required to build a base of supporting technologies and execute on potential quantum applications.[222]

Quantum workforce requirements are also subject to change as quantum technology progresses. A recent Government Accountability Office report notes that the proportion of quantum-specific to general STEM expertise required depends on the stage of technology development. While the fundamental quantum R&D process demands higher volumes of PhD-level talent, the commercialization process demands more generalized expertise.[223] Concerningly, "various indicators—including job board analyses and interviews with industry, academia, national laboratories, and government agencies—reflect that the United States maintains a 'significant unmet demand for talent at all levels.'"[224]

The general STEM skills gap in the United States is demonstrable and widely acknowledged.[225] The United States is "projected to face a shortfall of nearly 2 million STEM workers by 2025, due in part to insufficient processes to develop future STEM talent."[226] U.S. students historically perform poorly on national and international standardized STEM tests and maintain low STEM degree completion rates. They also face significant disparities in access to STEM resources based on socioeconomic status, race and ethnicity, and sex, resulting in the underrepresentation of minority groups in STEM careers.[227]

Access to talent will be a critical factor in determining which country prevails in quantum technology competition.

These trends present challenges for the quantum industry, which maintains a high demand for general STEM expertise. According to a QED-C survey assessing the workforce needs of 57 quantum companies, many of the employee aptitudes deemed most important to companies' success are broad STEM skill sets, rather than quantum-specific ones. Quantum companies need and struggle to find workers capable of software and algorithm development, analog and digital circuit design, and computer science and engineering. The quantum industry also struggles to retain these workers, whose broadly applicable skill sets are needed in a range of attractive technology sectors.[228]

The United States suffers quantum talent gaps at higher levels of expertise as well. According to a 2022 McKinsey & Company report, "Only about one qualified candidate was available for every three quantum-specific job openings, and more than half of quantum companies were actively hiring."[229] In the words of Yuval Boger, chief commercial officer at QuEra Computing, people with the high-level skills required to drive innovation in quantum computing are "unicorns" who "don't exist."[230] Recent research indicates that "less than 50 percent of quantum computing jobs will be filled by 2025 unless significant interventions occur."[231] This dearth of quantum-specific expertise may be due in part to a lack of quantum-specific curricula at earlier stages of schooling. According to the National Science and Technology Council, "exposure to Quantum Information Science and Technology (QIST) at the high school or undergraduate level in the United States is limited and rarely available prior to advanced coursework or research."[232] The "ensuing lack of awareness" about quantum often leads talented STEM students to "pursue other directions."[233]

Increased global interest in quantum technology further exacerbates the United States' shortage of quantum talent. The increased number of countries involved in quantum R&D contributes to a landscape in which "readily available quantum expertise is not only rare but globally dispersed," and countries must "compete to produce, attract, and retain top talent."[234] Most existing quantum-relevant talent "is educated and resides outside of the United States."[235] Only 10 of the top 20 quantum computing graduate degree programs are in the United States, a surprisingly low number given that the United States outspends every country but China on quantum technology R&D.[236] The highest number of "quantum-relevant graduates are educated in European Union, followed by India and China."[237] The European Union also has "the highest concentration of experts in quantum computing, followed by the United Kingdom and Russia."[238]

Even within the United States, the majority of quantum-relevant talent is foreign-born. According to an October 2021 report by the National Science and Technology Council, foreign talent constitutes approximately half of U.S. graduates in quantum-related fields. Further, dissertation records indicate that nearly half of U.S. PhDs focused directly on a quantum topic were awarded to foreign nationals between 2014 and 2018. Foreign talent plays a critical role in developing

U.S.-based quantum experts as well—the National Science and Technology Council noted that a number of the faculty members building quantum curricula and driving basic scientific research in the United States "are foreign-born themselves."[239] As such, international collaboration is key to both advance the quantum technology field and ensure U.S. competitiveness.

Several recent government initiatives aim to address the United States' shortage of quantum talent. The CHIPS Act directs the NSF to "evaluate the quantum workforce, integrate quantum into STEM curricula at all levels, and establish a 'Next Generation Quantum Leaders Pilot Program.'"[240] Similarly, the FY24 NDAA directs the DoD to establish a fellowship program for quantum technology students, with a particular focus on attracting "minority, geographically diverse, and economically disadvantaged" students who "historically lack access to quality STEM education resources and job opportunities."[241] And in August 2020, OSTP and NSF launched the National Q-12 Education Partnership between government, industry, professional societies, and the education community to enable "a foundation for classroom and curricula materials, developed in concert with key stakeholders from the quantum information science community."[242] But these efforts "may not be enough to give the United States the edge it needs to outcompete other countries that threaten to do better in producing and accessing quantum talent."[243]

To support the development of a robust quantum workforce, the U.S. government should:

Develop processes to understand workforce needs and skills gaps within the quantum ecosystem.

While there is broad consensus that the supply of quantum talent is insufficient to meet the demands of the National Quantum Strategy, the United States lacks a cohesive data source for determining workforce requirements and identifying skills gaps. As noted in a February 2022 National Science and Technology Council report, the United States needs "additional information, including but not limited to data on technical demands, training, awareness, and retention," to "comprehensively understand the long-term and short-term scope" of the quantum workforce shortage. More data on the range of skills needed to support quantum technologies would "better inform and track workforce development efforts."[244]

The Biden administration should direct the NQCO to conduct an annual systematic study of quantum workforce needs and projections. The study should identify current in-demand skills and anticipate which additional skills may be required in the subsequent year to push quantum research, development, and commercialization to the next level. The study should also detail the types of jobs available in the quantum ecosystem and which backgrounds are most relevant for those jobs. Such information would help job seekers identify and pursue suitable reskilling or upskilling opportunities. It would also help the academic community "make more informed decisions about how best to educate the future quantum workforce."[245] The NQCO should conduct the study in close consultation with the primary drivers of the NQI Act—NIST, NSF, and DOE—and the QED-C, which can provide additional insight about areas of concern.

The NQCO study should also assess the capacity of U.S. institutions—including government agencies, national labs, quantum companies, and the academic community—to effectively develop quantum talent. The study should identify existing workforce development and quantum education programs and generate metrics to assess their success and impact. Understanding which "initiatives work and why is critical to determine which programs can be replicated and scaled."[246]

Establish a national quantum education and workforce development center to harmonize and support existing initiatives to build a more robust pipeline of quantum talent.

The United States' current approach to quantum education and workforce development is highly decentralized, with government agencies, companies, and educational institutions each executing disparate programs and initiatives. While this approach is an important first step toward mitigating the United States' quantum workforce challenges, it can lead to disparities in the depth and scope of implementation efforts.[247]

Congress should include the establishment of a national quantum education and workforce development center in the NQI reauthorization to magnify existing workforce development efforts and streamline implementation.[248] The center would provide a venue for government, industry, and academia to connect ongoing initiatives, share resources and best practices, and brainstorm new programming. The center could develop and promote resources to help job seekers learn about the educational requirements of their desired careers, identify training opportunities, and connect with career counselors, mentors, and other job seekers.[249]

The center should also create a digital talent matching platform to connect job seekers to potential employers,

a suggestion similarly put forth in a previous Center for a New American Security (CNAS) report on the broader U.S. technology workforce.[250] The Tasmanian Government's Rapid Response Skills Matching Service, which helped businesses adversely affected by the COVID-19 pandemic fill critical vacancies, could serve as a model for this effort.[251] The platform should feature a centralized database to which job seekers can voluntarily upload a résumé, cover letter, relevant certificates, and related application materials. Government agencies, private quantum companies, and academic institutions could use the database to quickly identify candidates with the skills and experience required to fill key positions.[252]

Work with allies and partners to create cross-training and reciprocal research exchange programs with quantum centers worldwide.[253]

Quantum talent is a commodity in short supply both nationally and internationally, and every country involved in quantum R&D must contend with workforce challenges. No country has exclusive access to the people required to advance state-of-the-art quantum technologies.[254] The Biden administration can therefore collaborate with allies and partners to mitigate key skills gaps and create opportunities for every country to benefit from the quantum expertise that is readily available.

The Biden administration can start by expanding the scope and participation of existing multilateral workforce development efforts. For example, the Biden administration should encourage the Quad partners to increase the size of the Quad Fellowship program, which aims to develop a network of STEM talent by sponsoring American, Australian, Indian, and Japanese master's and doctoral students to study in the United States.[255] The Fellowship currently supports 100 students each year. Opening the fellowship to more students, particularly students in quantum-relevant disciplines, such as physics, electrical engineering, and computer and information science, will enable the Quad to stay at the cutting edge of research and discovery and counteract competitors' efforts to train, recruit, and retain the best talent. The Quad should also create opportunities for STEM fellows to study in Australia, Japan, and India, as geographically diverse fellowship placement options will help ensure that all Quad partners benefit from the exchange of STEM talent and could help broaden the pool of potential talent from which to draw.[256]

The budding trilateral relationship between the United States, South Korea, and Japan presents another opportunity for international cooperation. The United States already has bilateral government-to-government agreements with both Japan and South Korea that underscore the importance of quantum workforce development.[257] The United States also has academic and industry partnerships with both countries, including an IBM and Google-supported partnership between the University of Chicago and the University of Tokyo, and a memorandum of understanding between IBM, IonQ, and the Korean government. The OSTP should develop metrics to assess the effectiveness of these existing initiatives and scale those that are effective into trilateral frameworks.[258]

Universities and quantum companies also have important roles to play in improving the United States' access to quantum-specific and general STEM talent. To improve educational opportunities, increase interest in quantum careers, and promote diversity, equity, and inclusion in the quantum industry, universities and private industry should:

Cooperate to develop industry-specific postsecondary education programs and create a pipeline of candidates ready to fill critical gaps.

As noted in a previous CNAS report on the U.S. technology workforce, academic institutions and industry must work together to improve quantum education and proficiency.[259] Purdue University's Semiconductor Degrees and Credentials Program, which combines classroom lectures with experiential learning opportunities, is one example of academic-industry cooperation that can be modeled and expanded within the quantum ecosystem.

The Purdue program gives undergraduate and graduate students a variety of pathways—including a master's degree, undergraduate minors, associate degrees, and stackable certificates—to gain core skills needed for the semiconductor industry.[260] The Semiconductor Degrees Leadership Board, comprised of private sector executives, plays a critical role in crafting the program's curricula based on industry requirements.[261] Some industry partners, such as semiconductor manufacturer SkyWater Technology, even administer the program's introductory seminar. Industry partners also agreed to offer cooperative and internship opportunities to Purdue students, enabling them to gain hands-on experience while completing their degrees.[262] With over 11,000 undergraduate and 4,000 graduate students, Purdue's program helps maintain a pipeline of globally competitive talent.[263]

Leading quantum companies like IBM, IonQ, and PsiQuantum could similarly partner with colleges and universities to bolster and expand quantum computing degree programs and ensure that students gain the practical experience required to fill critical talent gaps immediately upon graduation.[264] Industry partners can also help increase the attractiveness of quantum degree programs by funding scholarships, agreeing to interview students that maintain a certain grade point average, and providing employees with competitive student loan repayment benefits. Such efforts would help lower-income students pursue quantum degrees and increase the likelihood that graduates ultimately use their degrees to pursue careers in the quantum field.

Industry could further support university programs by hosting faculty at their manufacturing, research, and development facilities for information sessions, tours, and roundtable discussions. Increased communication and cooperation between industry leaders and university faculty would help the two communities better understand each other's needs and facilitate resource sharing.[265]

Partner with K–12 school systems to develop quantum curricula and experiential learning opportunities for students.

The National Q-12 Education Partnership offers a framework for integrating quantum education into existing STEM content and curricula at the K–12 level.[266] As Emily Edwards noted in recent congressional testimony, however, "there is still significant work to be done on curriculum development and implementation" and K–12 teachers "need more scaffolding to demystify quantum and build their confidence in teaching the topic."[267] Initiatives like George Mason University's (GMU's) K–12 Quantum Workforce Development Project can help equip K–12 schools with the resources required to increase students' exposure to and understanding of quantum.

The K–12 Quantum Workforce Development Project, which is coordinated by faculty from GMU's Quantum Science and Engineering Center and supported by federal funding, aims to "inspire the next generation of students to pursue quantum careers and simultaneously prepare a diverse quantum workforce in Northern Virginia."[268] As part of the project, GMU faculty run workshops for K–12 teachers in Loudoun and Fairfax counties to "develop teachers' quantum knowledge, work with teachers to adapt existing materials, and develop new materials to integrate into K–12 classrooms." The project also provides stipends for elementary and high school teachers to attend professional development seminars that give teachers the tools to integrate quantum fundamentals into existing curricula.

GMU hosts a free summer immersion program from rising high school seniors as well, which "supports student learning of quantum concepts and quantum careers." The program allows students to "complete a virtual learning program and then spend a week living on campus and visiting local quantum laboratories, industries, and government agencies" working in quantum.[269] Students have had the opportunity to meet quantum experts at NASA, NIST, and MITRE Labs, among other organizations, as well as undergraduate and graduate GMU students studying quantum science. According to the Virginia Economic Development Partnership, the project has helped accelerate local K–12 schools' adoption of quantum curricula, increase students' interest in quantum careers, and build new skills to support quantum-focused organizations.[270]

Universities with quantum degree programs and faculty with quantum expertise—such as the Massachusetts Institute of Technology, the University of Chicago, and the University of California, Berkeley—should launch similar initiatives in other states. The quantum industry can support such initiatives by offering to host K–12 teachers and students at their R&D facilities, clearly communicating their anticipated workforce requirements, and promoting their achievements to generate excitement about careers in quantum.

Partner with state and local governments to expand quantum education and professional development opportunities to individuals in rural areas.

A previous CNAS report on the U.S. technology workforce found that projected growth in the STEM field is expected to disproportionately benefit workers residing in metropolitan areas in coastal states, who typically hold advanced degrees.[271] But individuals in rural areas, as well as individuals who lack college degrees, comprise a sizable untapped talent pool in the United States, and the quantum industry is well-positioned to offer them stable, high-paying jobs.[272]

To fill critical talent gaps, quantum companies should design reskilling and upskilling initiatives to attract and uplift these communities. Quantum apprenticeship programs, accelerated skill development programs, or online certification programs would allow such underrepresented individuals to hone their skills and advance their earning and promotion potential.

Quantum companies could also host job fairs in rural communities to stimulate interest in the quantum industry, offer education opportunities, and allow residents to learn about job openings. They could hold seminars or briefings from quantum industry representatives, information sessions on pathways to quantum careers, and panel discussions about how to apply for relevant professional development opportunities. Such events should be well-advertised in local grocery stores, gas stations, libraries, and schools and held at various times of day, allowing both shift and traditional workers to attend.[273]

Conclusion

QIS is a critical and emerging technology area that could help shape geopolitics in the 21st century. Quantum computing is a particularly influential subset of QIS that promises to confer tremendous economic and military benefits by dramatically improving computational speed and processing power across a range of industries. The country with the best quantum computers may unlock advantages in several key areas, including pharmaceuticals, batteries, fertilizers, intelligence collection, and advanced weapons systems. Given the stakes, quantum computing has become a central front in U.S.-China strategic competition. Maintaining America's quantum computing edge is therefore critical to U.S. national and economic security.

Thanks to strong government support, a robust and innovative private industry, and a high-quality research infrastructure, the United States is the global leader in many approaches to quantum computing, including neutral atoms, trapped ions, and photonics. But China is catching up. Chinese researchers and engineers have reached rough parity with their U.S. counterparts in superconducting qubits. They have also unlocked several significant breakthroughs in other QIS subsets that could expedite their development of highly capable quantum computers.

The CCP has pushed quantum computing to the top of its list of strategic technology priorities. Chinese government officials have bluntly stated their intent to beat the United States and establish "supremacy" in quantum by building an independent manufacturing chain, nurturing quantum talent, and developing market-ready applications.[274] This heightened government attention to and investment in quantum will not guarantee success—China still maintains several weaknesses that will be challenging to overcome, such as the relatively small size and scale of its quantum industry. Nonetheless, Chinese technology companies have a track record of achieving rapid and unexpected technical breakthroughs following periods of government intervention in R&D processes.[275] U.S. leadership in quantum computing is thus far from assured.

Should China take the lead in quantum computing, it would gain a powerful new toolkit of capabilities that could be used to "threaten previously unreachable parts of the United States' corporate, military, and government infrastructure."[276] China's development of a powerful quantum computer would allow it to crack standard encryption protocols and expose sensitive intelligence, as well as Americans' personal data. China would gain an upper hand in establishing market dominance as well, making it more difficult for U.S. quantum companies to compete internationally. It could also establish quantum technology governance frameworks that exacerbate global socioeconomic and ethnic divisions and undermine democratic principles.

Perhaps most concerning, China may not need to achieve global quantum leadership to introduce such national and economic security threats. Many quantum computing applications will have long development and deployment timelines that are not materially affected by a one- or two-year capability gap. In other words, even if China remains a few years behind the United States in quantum R&D, it can accelerate progress toward problematic military and commercial end uses so long as it continues developing a domestic infrastructure that supports quantum implementation. The level of threat China poses in quantum technology does not depend solely on its position relative to the United States, but also on its ability to quickly infuse the computational advantage afforded by quantum computing into an overarching production and deployment process.

This means that the best way for the United States to stay ahead in quantum computing is to emphasize policies that accelerate U.S. progress at a rate unattainable by select adversaries. U.S. policymakers can shape a brighter future by acting quickly to promote the United States' quantum advantages and ensure the development of a global quantum ecosystem that harnesses the technology's positive potential and mitigates its risk. In the near term, policymakers should focus on supporting the development of a resilient quantum technology supply chain and building a robust and diverse quantum workforce.

The quantum technology supply chain is inherently risky—it is highly dispersed and specialized with many single points of failure. At the same time, however, it is nimble, presenting numerous opportunities for U.S.

policymakers to mitigate vulnerabilities and promote U.S. interests. The White House can support the development of a resilient supply chain by conducting a comprehensive supply chain review and leveraging financial incentives, as well as the Tech Hubs Program, to promote domestic production of sensitive quantum technology inputs. The White House should also equip the QED-C to quickly detect new supply chain vulnerabilities as quantum technologies evolve, and establish an international forum to coordinate supply chains with U.S. allies and partners. These measures will give policymakers the information required to actively choose where to accept risk, rather than passively assuming risk due to a lack of awareness and understanding.

A well-trained workforce is also critical to realizing and maximizing the full potential of quantum computing. Talent is required not only to invent, develop, and scale quantum technologies but also to identify quantum applications and ensure responsible technology use. The United States lacks both the quantum-specific and general STEM expertise needed to continue driving quantum technologies forward. The U.S. government can foster a more robust talent pipeline by developing processes to understand workforce needs and skills gaps within the quantum ecosystem, establishing a national quantum education and workforce development center, and creating research exchange programs with allies and partners. Industry and academia can support these initiatives by creating industry-specific postsecondary education programs, partnering with local school systems to build quantum curricula and experiential learning opportunities for K–12 students, and expanding quantum education and professional development programs to individuals who are historically underrepresented in STEM professions, such as those who reside in rural areas or lack college degrees. These efforts will help the United States outcompete other countries also striving to produce and access quantum talent.

When leaving his post in March 2022, former DoD Chief Data Officer David Spirk warned that there were "not enough senior leaders getting their heads around the implications of quantum."[277] Spirk called on policymakers to "speed up" their efforts to prepare for the quantum future or risk falling behind adversaries determined to produce the most advanced technologies.[278] As Spirk's comments indicate, the United States is at an important inflection point. Quantum computing is maturing rapidly, and researchers and engineers across the globe are at the precipice of significant technological breakthroughs. But actual applications remain limited and the path to creating quantum systems of significant scale is uncertain. Now is the time to formulate policies that safeguard the United States' quantum advantages. Policymakers should heed Spirk's call to "pick up the pace" on quantum.[279]

U.S. policymakers can shape a brighter future by acting quickly to promote the United States' quantum advantages and ensure the development of a global quantum ecosystem that harnesses the technology's positive potential and mitigates its risk.

Appendix: Appendix: Global Entities Engaged in Quantum R&D

Below is a non-exhaustive list of the companies and institutions in China, as well as U.S.-allied countries, engaged in material R&D efforts pertaining to quantum computing hardware. The information is extracted from data collected by Global Quantum Intelligence.[280]

Company	Country	Modality
Archer Materials Limited	Australia	Spin/Quantum Dot/ Complementary Metal-Oxide-Semiconductor ("CMOS")
Centre for Quantum Computation & Communication Technology (CQC2T)	Australia	Spin/Quantum Dot/CMOS and Photonic
Diraq	Australia	Spin/Quantum Dot/CMOS
Griffith University	Australia	Photonic
University of Queensland	Australia	Photonic
Quantum Brilliance	Australia	Nitrogen Vacancy Center ("NV")
Silicon Quantum Computing	Australia	Spin/Quantum Dot/CMOS
Alpine Quantum Technologies	Austria	Trapped Ion
Parity Quantum Computing	Austria	Trapped Ion
University of Innsbruck	Austria	Trapped Ion
NTT	Austria	NV
IMEC	Belgium	Superconducting
Anyon	Canada	Superconducting
D-Wave	Canada	Annealing and Gate Models
InfinityQ	Canada	Spin/Quantum Dot/CMOS
Nord Quantique	Canada	Superconducting
Photonic	Canada	Spin/Quantum Dot/CMOS and Photonic
Simon Fraser University	Canada	Spin/Quantum Dot/CMOS and Photonic
University of Waterloo	Canada	Superconducting/Trapped Ion/ Photonic
Xanadu	Canada	Photonic
Bose Quantum	China	Photonic
Origin Quantum Computing	China	Superconducting and Spin/ Quantum Dot/CMOS
Qike Quantum	China	Trapped Ion
SpinQ	China	Nuclear Magnetic Resonance ("NMR")

Company	Country	Modality
TuringQ	China	Photonic
University of Science & Technology of China (USTC)	China	Superconducting/Photonic
Wuhan Institute of Physics & Mathematics, Chinese Academy of Sciences	China	Cold/Neutral/Helium Atom
Niels Bohr Institute	Denmark	Sping/Quantum Dot/CMOS and Superconducting
Sparrow Quantum	Denmark	Photonic
Technical University of Denmark	Denmark	Photonic
Aalto University	Finland	Superconducting
IQM Finland	Finland	Superconducting
Nokia Bell Labs	Finland	Topological
Alice&Bob	France	Superconducting
C12 Quantum Electronics	France	Spin/Quantum Dot/CMOS
CEA-Leti/Inac	France	Spin/Quantum Dot/CMOS
Institut d'Optique	France	Cold/Neutral/Helium Atom
NextGenQ	France	Trapped Ion
Pasqal	France	Cold/Neutral/Helium Atom
Quandela	France	Photonic
Siquance	France	Spin/Quantum Dot/CMOS
ATIQ Project	Germany	Trapped Ion
eleQtron	Germany	Trapped Ion
Infineon	Germany	Superconducting, Trapped Ion, and Spin/Quantum Dot/CMOS
Planqc	Germany	Cold/Neutral/Helium Atom
Q.ANT	Germany	Photonic
Quantum Factory	Germany	Trapped Ion
QUDORA Technologies	Germany	Superconducting
RWTH Aachen	Germany	Spin/Quantum Dot/CMOS

Company	Country	Modality
University of Duisburg-Essen	Germany	NV
University of Konstanz	Germany	Spin/Quantum Dot/CMOS
University of Stuttgart (PhotonQ Project)	Germany	Photonic
QpiAI	India	Superconducting
Silicofeller	India	Superconducting
Tata Institute of Fundamental Research	India	Superconducting
Equal1	Ireland	Spin/Quantum Dot/CMOS
Quantum Source Labs	Israel	Photonic
Weizmann Institute	Israel	Trapped Ion
Rotonium	Italy	Photonic
Fujitsu	Japan	Superconducting and NV Diamond
Hitachi Cambridge Laboratory	Japan	Spin/Quantum Dot/CMOS
NEC	Japan	Adiabatic/Annealing
The Nippon Telegraph and Telephone Corporation (NTT)/National Institute of Informatics (NII)/University of Tokyo	Japan	Photonic
Riken	Japan	Spin/Quantum Dot/CMOS
Riken/Fujitsu	Japan	Superconducting
Toshiba	Japan	Superconducting
TU Wien/NII Tokyo	Japan	NV
Korea Institute of Science & Technology	Korea	NV
NXP Semiconductors	Netherlands	Trapped Ion
Quantware	Netherlands	Superconducting
QuiX	Netherlands	Photonic
QuTech	Netherlands	Superconducting, Spin/Quantum Dot/CMOS, NV, Topological
Qilimanjaro	Spain	Adiabatic/Annealing
Quantic	Spain	Superconducting
Chalmers University of Technology	Sweden	Superconducting and Topological
Stockholm University	Sweden	Trapped Ion
ETH Zurich/Paul Sherrer Institute	Switzerland	Superconducting and Trapped Ion
ID Quantique	Switzerland	Photonic
University of Basel	Switzerland	Spin/Quantum Dot/CMOS

Company	Country	Modality
Hon Hai (Foxconn)	Taiwan	Trapped Ion
Technology Innovation Institute	United Arab Emirates	Superconducting
Networked Quantum Information Technologies (NQIT)	United Kingdom	Trapped Ion
ORCA Computing	United Kingdom	Photonic
Oxford Quantum Circuits	United Kingdom	Superconducting
Oxford University	United Kingdom	Superconducting, Trapped Ion, Photonic, and NV
Quantum Motion Technologies	United Kingdom	Spin/Quantum Dot/CMOS
TundraSystems Global	United Kingdom	Photonic
Universal Quantum	United Kingdom	Trapped Ion
University of Bristol	United Kingdom	Photonic
University of Sussex	United Kingdom	Trapped Ion

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